Global competition for economic migration

Visa menus, worker histories and the allocation of talent

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The economic question

Governments offer access, costs and rights

Admission is one part of a package that shapes a person's future options

Access to employment

- Occupation, citizenship and sponsorship can determine which routes a worker can use.
- Employer portability changes the jobs available **after admission**.
- A route opening can redirect workers from another destination rather than create a new move.

Residence and membership

- Family work rights, renewal and settlement change the value of the initial offer.
- Argentina's investment-linked **citizenship** initiative illustrates a different membership instrument—not a work visa.
- Announced terms, binding rules and operational access are distinct.

How do these offers change the allocation of workers—and the offers made by rival governments?

▶ Legal examples ▶ Nationality-specific routes

How do governments respond to workers and rivals?

Worker substitution connects visa menus to competition for talent

Government offers → worker choices and composition → rival offers

- **Worker side:** which terms affect participation, destination, employer choice and retention?
- **Government side:** do countries compete through fees, eligibility and rights?
- **Firm side:** how do recruitment, sponsorship and employer dependence affect wages and allocation?

Which allocation and welfare consequences follow from a change in one country's offer?

> Measure menus and careers first. Estimate worker substitution, then add the decision problem of firms or governments.

Connecting legal offers with global career allocation

The contribution combines route-level policy bundles with worker histories

Closest literature	What it establishes	What this project adds
Measuring policy — IMPIC; DEMIG; high-skilled policies (Helbling et al. 2017; de Haas et al. 2018; Czaika and Parsons 2017)	Legal levels, policy changes and instruments differ from realised access.	Dated **joint-route bundles**, broad migration motives and occupation–education comparisons, with explicit coverage and unresolved rules.
International allocation (Bertoli and Fernández-Huertas Moraga 2013; Brinatti and Guo 2024)	Competing destinations and third-country diversion matter.	Connect legal-menu variation to **domestic and international career histories**, not only bilateral totals or one policy corridor.
Legal options and labour markets (Townsend and Allan 2025; Borusyak et al. 2026)	Employer restrictions and connected alternatives affect mobility and equilibrium responses.	Study **cross-country offers** as inputs to government competition; add employer responses as a distinct extension.

Measurement and descriptive evidence are available; strategic policy and welfare require further identification.

▶ Full literature map ▶ Demand identification ▶ Government objectives

Data: offers and histories

One Atlas, two layers: inventory and characteristics

The retained delivery distinguishes migration motives from detailed contract terms

Broad pathway inventory

5,744 country/name entries · 194 destination labels

Work, study, family, protection, business/investment, visit and other categories.

Search-window metadata: **2000–2025**.

This is a retained inventory, not certified exhaustive legal discovery.

Detailed annual characteristics

2,195 country/route IDs · 43,049 route-years

80 value fields + 80 status fields, with four identifying columns.

Work-oriented selection precedes detailed extraction. Native unit: **destination × route × year**.

Eligibility · financial burdens · procedures · employment rights · family rights · renewal and settlement

▶ Categories and fields ▶ Identity and history

Revelio follows domestic and international job histories

People → positions → ordered histories → domestic and international job transitions

Retained foundation	Broad verified-geography sample, 2010–2025
549.7 million keyed people	**892.2 million** domestic transitions
1.766 billion positions	**83.65 million** international transitions

One person can contribute several different events

Domestic employer change → international job-country change → return or onward transition

- These are **career events**, not counts of visa grants or necessarily residential moves.
- The strict whole-history sample retains **303.5 million people**; its denominator differs from the broad sample above.

▶ Sample and chronology ▶ Gaps and attrition ▶ Horizon outcomes

Matching policies to careers requires comparable profiles

Legal eligibility and observed career characteristics are not the same object

Dimension	Atlas and policy comparison	Revelio and analytical support
Time	2000–2025 presentation window	Verified transition comparison: 2010–2025
Occupation	22 nonmilitary SOC major groups	Finer O*NET codes retained; SOC major groups used here
Origin	Citizenship/residence conditions	Prior recorded job country—not citizenship
Education	Route-specific requirements	Lifetime highest degree need not precede a move

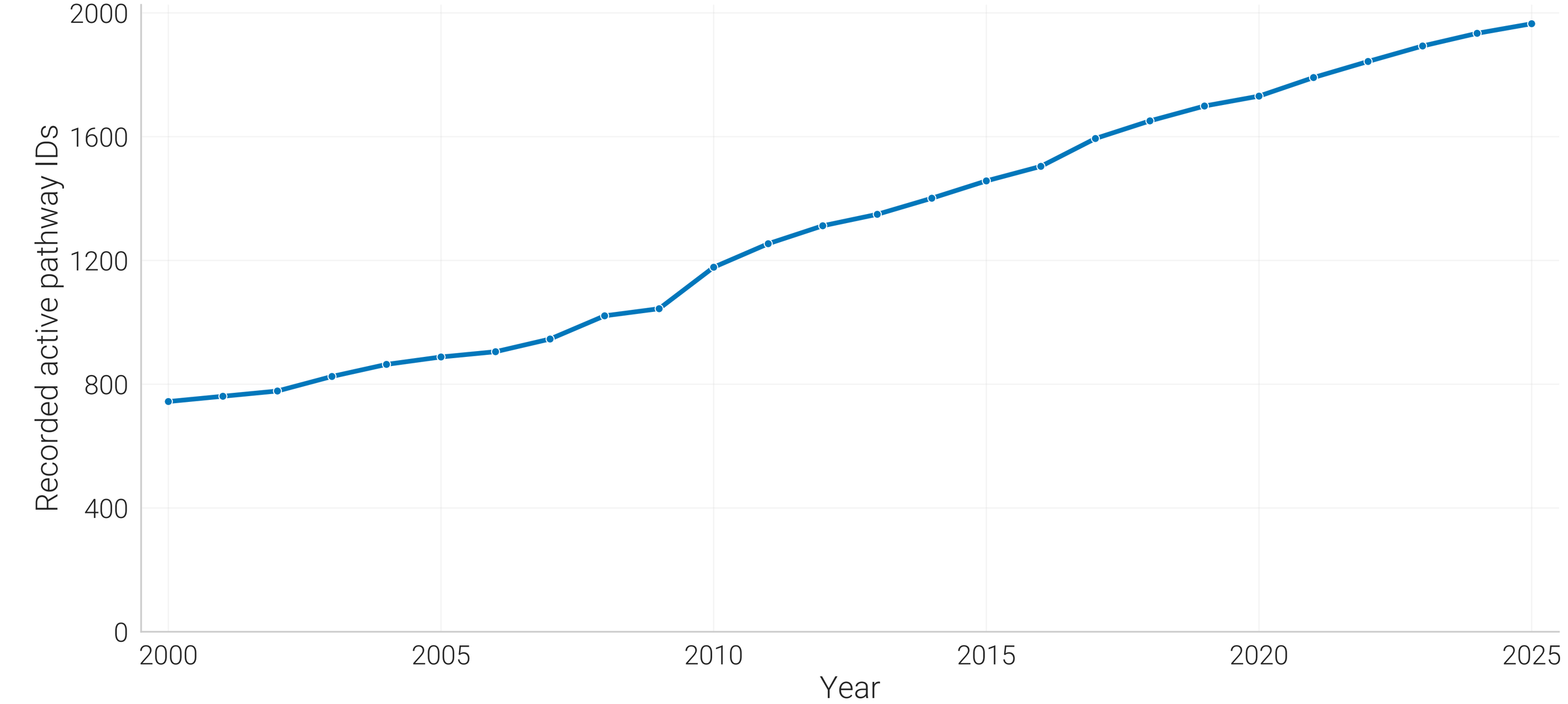
Passing occupation and education screens is **partial compatibility**, not complete admission eligibility.

▶ All occupational levels ▶ Coverage ▶ Unresolved rules

Three descriptive facts

Recorded active pathways increase over time

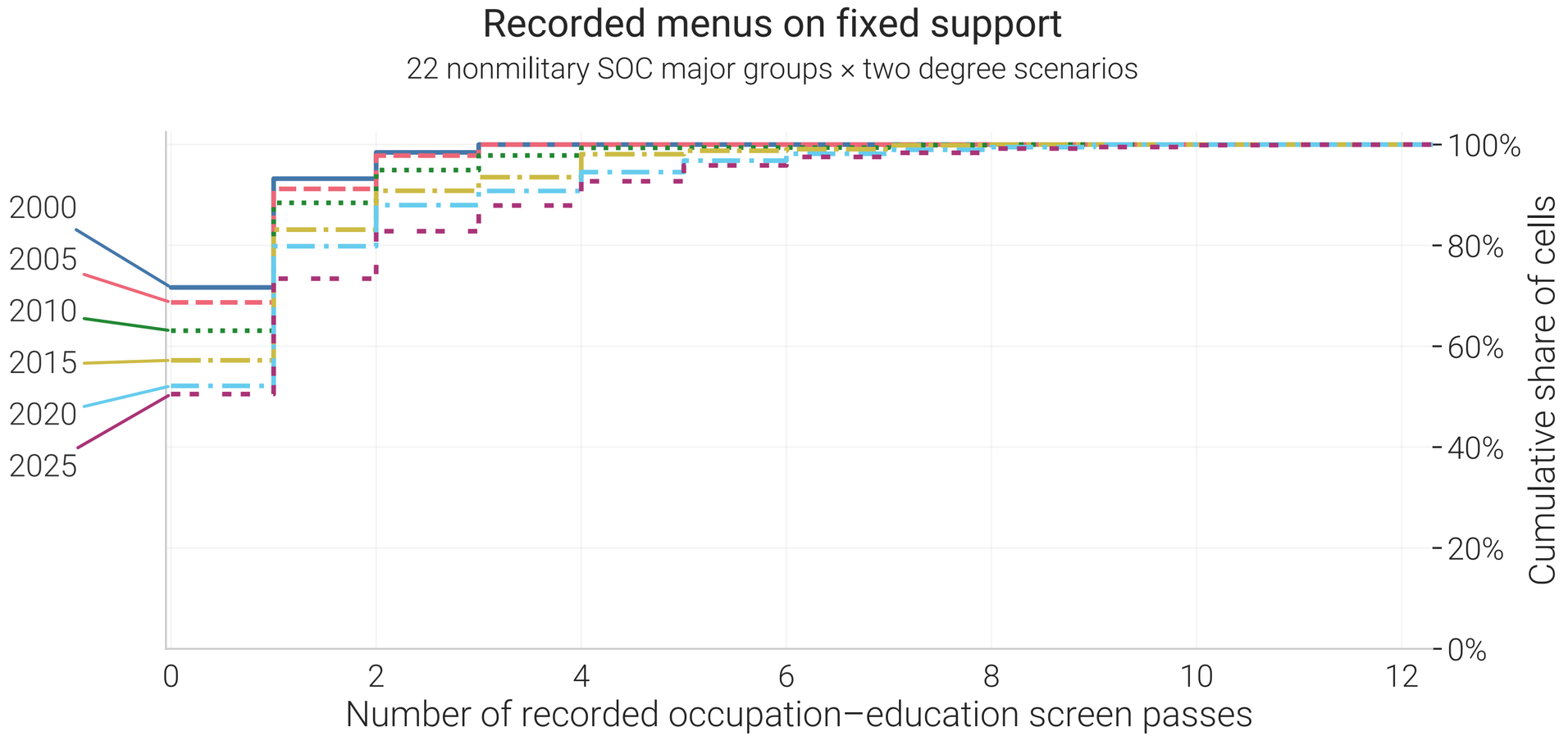
Work-oriented route IDs · 2000–2025 · 744 → 1,965



▶ Country representation ▶ Appearances and endings

Menu expansion survives fixed destination–profile support

167 destinations × 44 occupation–education scenarios · equal cell weights · six dates



Fixed 167 destinations; equal cell weights. Exact ECDF; no smoothing.

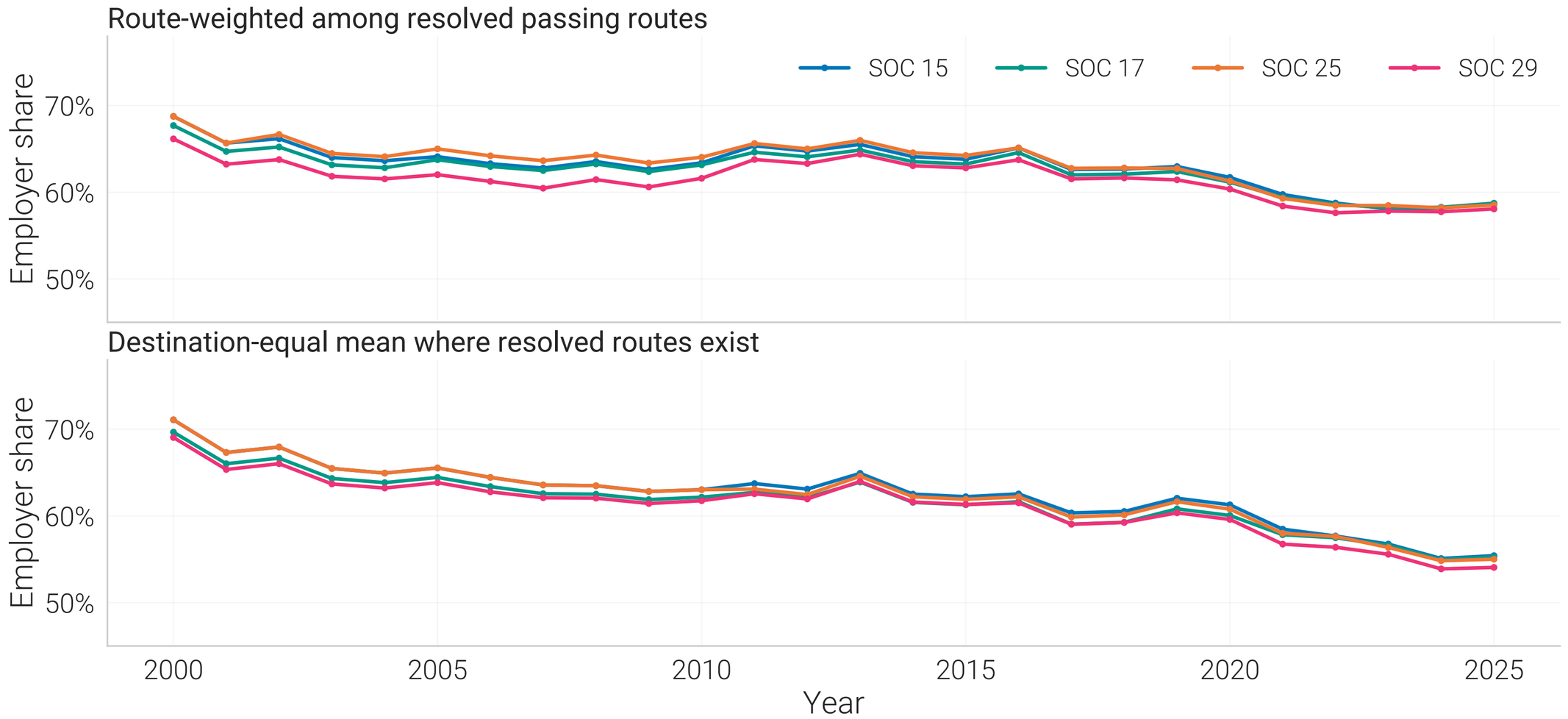
Zero coded passes are retained. Unknown screens are not zero legal access; nationality is not evaluated.

Retained hypothetical scenarios overlap; these are neither people nor complete legal choice sets.

▶ Changing support ▶ Computer/math example ▶ Healthcare example

Employer sponsorship remains common as its recorded share falls

2000–2025 · four occupation profiles · undergraduate screen · resolved sponsorship



▶ Weights ▶ Education rules ▶ Family rights ▶ Fee contrasts

A profile can face different joint requirement bundles

Czechia, 2024 · SOC 15 undergraduate screen · three of 12 passing route IDs

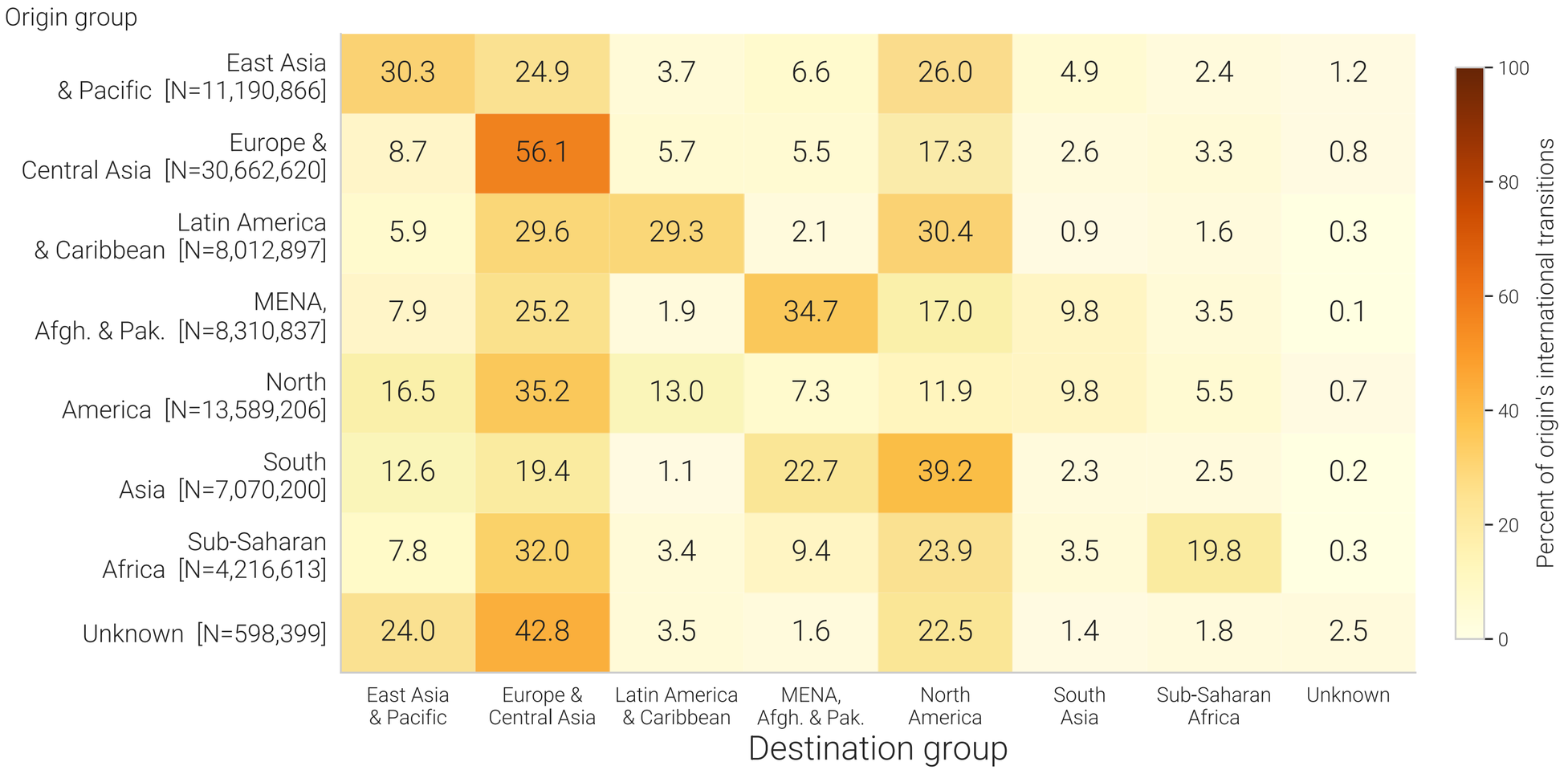
Recorded route	Sponsor	Applicant fee	Annual salary floor	Nationality condition
Digital Nomad Visa	None	CZK 2,500	CZK 780,144	AU, CA, GB, JP, KR, NZ, TW, US
Employee Card	Employer	CZK 7,500	CZK 226,800	No recorded gate
EU Blue Card	Employer	CZK 5,000	CZK 780,138	Unknown

- Each row is an **actual recorded package**, not independent field minima.
- Occupation/education compatibility is not complete eligibility for one person.
- Nationality, admission risk and continuation rights prevent an unqualified dominance claim.

▶ Dominance test ▶ Preferential routes

International transitions between regions

2010–2025 · fixed World Bank 2026 regions · origin-conditional shares

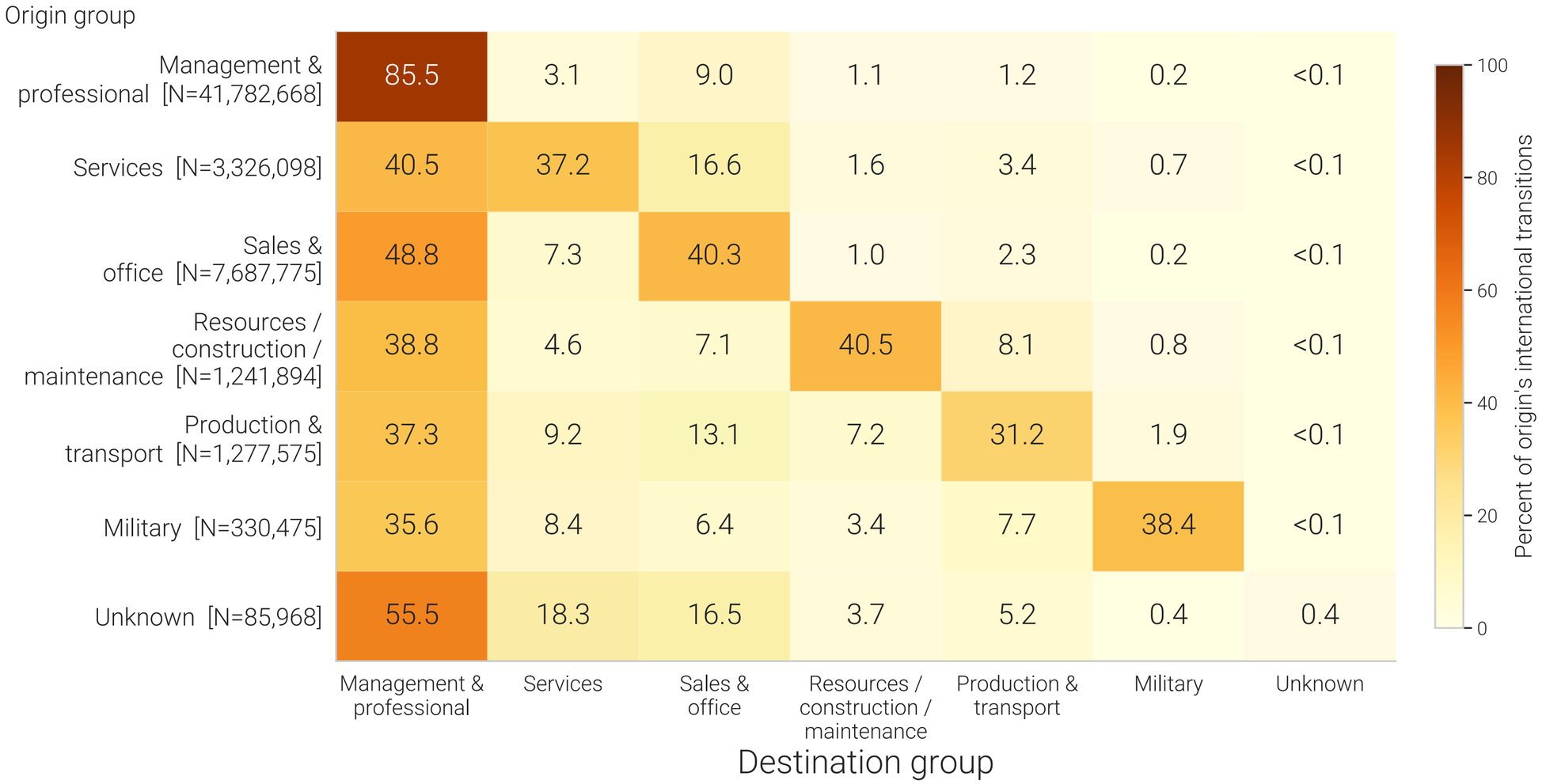


83.65m international job transitions · broad verified-country sample · unknown groups retained

▶ Region counts ▶ Income-group transitions ▶ Income counts

International transitions between occupational families

2010–2025 · six analytical SOC families plus unknown · not a skill ladder



55.73m international job transitions · strict histories · unknown included · families are not skill ranks

▶ Family counts ▶ All 23 SOC groups ▶ Domestic comparison ▶ Skill/task interpretation

Worker responses and policy incentives

Worker substitution links legal offers to allocation

First distinguish whether to move from where to go conditional on moving

$$F_{rd} = N_r \, m_r \, s_{rd}, \qquad p_{rd} = \frac{\exp(V_{rd})}{\sum_{j \in A_r} \exp(V_{rj})}.$$

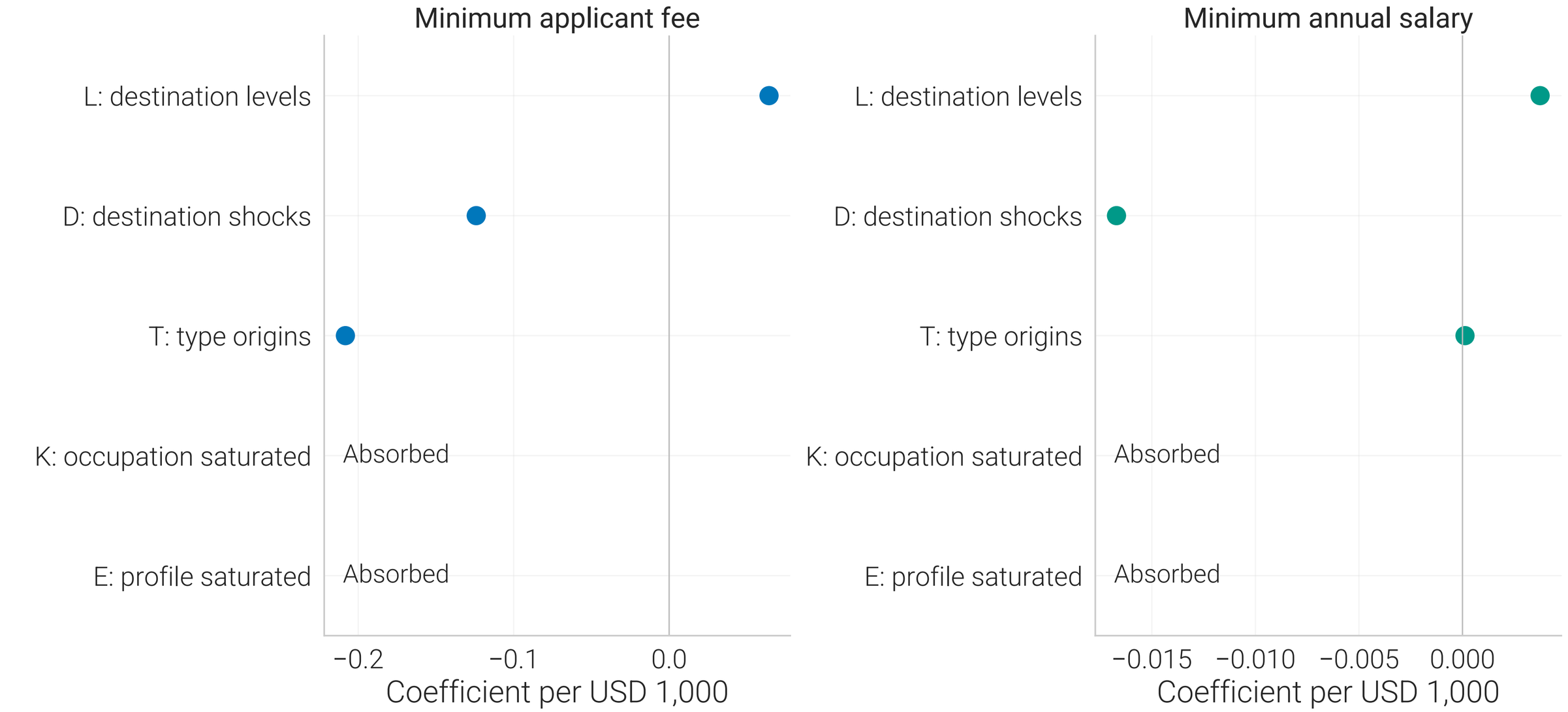
- N_r: population at risk; m_r: participation; s_{rd}: share of movers choosing d.
- A_r must describe the relevant alternatives, including staying when incidence is the question.
- Current positive-cell international fits describe **conditional recorded allocation**, not all migration participation.

Which fee, rights or eligibility change identifies substitution—and which variation survives the fixed effects?

▶ Denominators and MPML ▶ Fixed effects ▶ Preference versus access

Preliminary monetary coefficients depend on the retained variation

Separate-policy conditional-index slopes · per USD 1,000 · no current standard errors



From legal offers to competition for talent

Worker substitution is an input—not the whole government or firm model

What the evidence establishes

Recorded menus expand on fixed support; routes carry different joint terms; career transitions cross countries and occupations.

What must be added

- **Worker responses:** usable access, a defined outside option and credible policy variation.
- **Government or firm choices:** objectives, capacities, feasible deviations and payoff-relevant composition.
- **Welfare:** which people count, market adjustment, future rights and the information path.

Counterfactual: how would workers and rival offers change if one country relaxed a specified route's portability or fee?

▶ Government cost inversion ▶ Welfare scope ▶ All references

Appendix: legal objects and data

Admission, residence and citizenship are distinct

Example	Institutional distinction	Presentation scope
Argentina investment-linked citizenship	2025 decrees create category/procedure; October 2026 announcement describes financial/family terms.	Citizenship, not a work visa. Exact operational opening not established.
Citizenship-conditioned work access	A potential migrant's nationality can determine a preferential route.	Previous job country cannot assign nationality.
Employer-tied work permission	Admission may leave the holder dependent on a sponsor.	Portability and continuation can matter without an immediate move.

The US H-1B1 comparison and Germany's Western Balkans employment regulation provide source-backed nationality examples; they are not a verified identification of the separately mentioned recent German occupation-specific announcement.

▶ Return: motivation

A nationality preference need not dominate a general route

- H-1B1 for Chile/Singapore has nationality-specific allocations and a distinct visa-abroad procedure.
- Temporary-intent and duration/renewal terms differ from the ordinary H-1B framework.
- Germany’s Western Balkans route illustrates nationality-conditioned access with different qualification rules.

Compare the whole bundle for the same applicant and date: eligibility, sponsor, fees, intent, work rights and continuation.

▶ Return: motivation ▶ Return: route bundles

The broad inventory includes more than work routes

Retained category code	Entries
Work	1,480
Study	446
Family	571
Protection	576
Business/investment	579
Visit	669
Other	1,423
Total	**5,744**

Characteristic families

- Access and selection
- Fees, funds, salary and investment
- Procedures and compliance
- Work rights and portability
- Family and dependent rights
- Duration, renewal and settlement
- Identity, status and timing

The detailed release has **80 values and 80 status companions**, not 80 independent screening instruments. A separate AI-category field is blank in 3,693 inventory entries.

▶ Return: Atlas

Route identity and annual history require validation

- ID: country plus supplied/slugified route name—not an adjudicated legal equivalence class.
- Annual states use lifecycle bounds, recorded changes and forward-filling; they are not independent annual legal observations.
- Renaming, branching and succession can change counts without adding a simultaneous option.
- Unknown, inapplicable, uncertain and known values must remain distinct.

Raw annual years: 1940–2026; intended search/display window: 2000–2025. The UK has only two IDs in the unchanged 2024 retained panel, a consequential coverage warning.

▶ Return: Atlas ▶ Return: growth

Occupational detail and analytical restrictions differ

Level or selection	Count/scope	Interpretation
Official SOC major/minor/broad/detailed	23 / 98 / 459 / 867	Reference taxonomy, not observed vendor counts
Atlas major-group scenarios	22	Military excluded; national classifications mapped into SOC
Earlier skilled-group analysis	11	Analytical restriction
Current monetary gravity	15, 17, 25, 29	Four selected major groups
Retained Revelio source	O*NET code; older `role_k1500` fallback	Fine counts, complete vendor hierarchy and edition unverified

The observed occupation source includes military and malformed values. Exact recognized-code rules differ from simply taking two initial characters.

▶ Return: bridge ▶ Return: occupations

History construction changes the retained population

Object	People	Positions/events
Full keyed foundation	549,729,026	1,765,970,337 positions
Strict whole histories	303,499,990	1,117,773,296 positions
Strict adjacent events, all retained years	—	814,273,306 events
Broad verified geography, 2010–2025	—	975,853,764 events

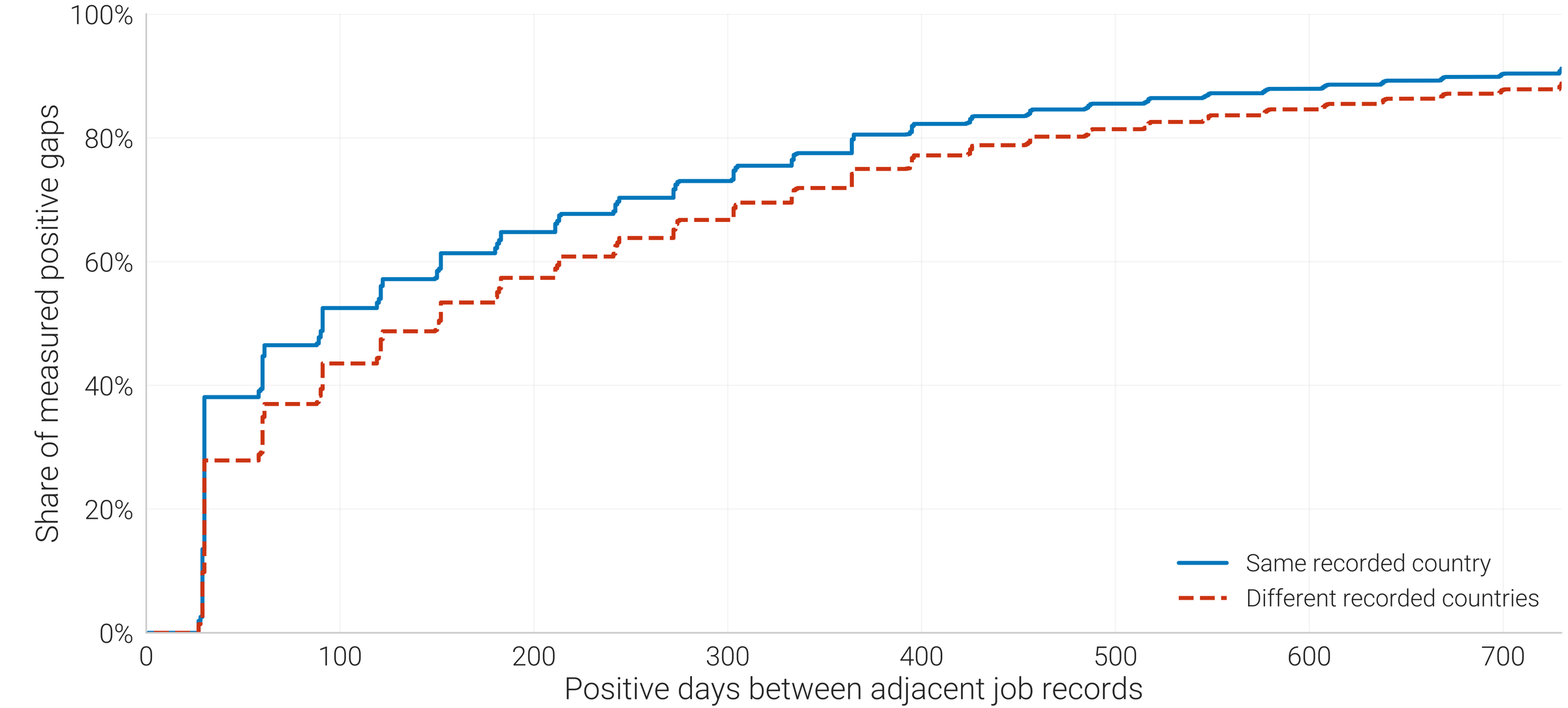
Missing starts, countries and tied starts have **overlapping** exclusion counts. The strict rule excludes a person if any history record fails; the broad geography construction removes missing-country records before ordering.

These are different populations, not successive rows in one funnel. Concurrent positions, missing ends and chronology cannot be resolved by arbitrary sorting.

▶ Return: histories

Many positive gaps are short; missing updates still matter

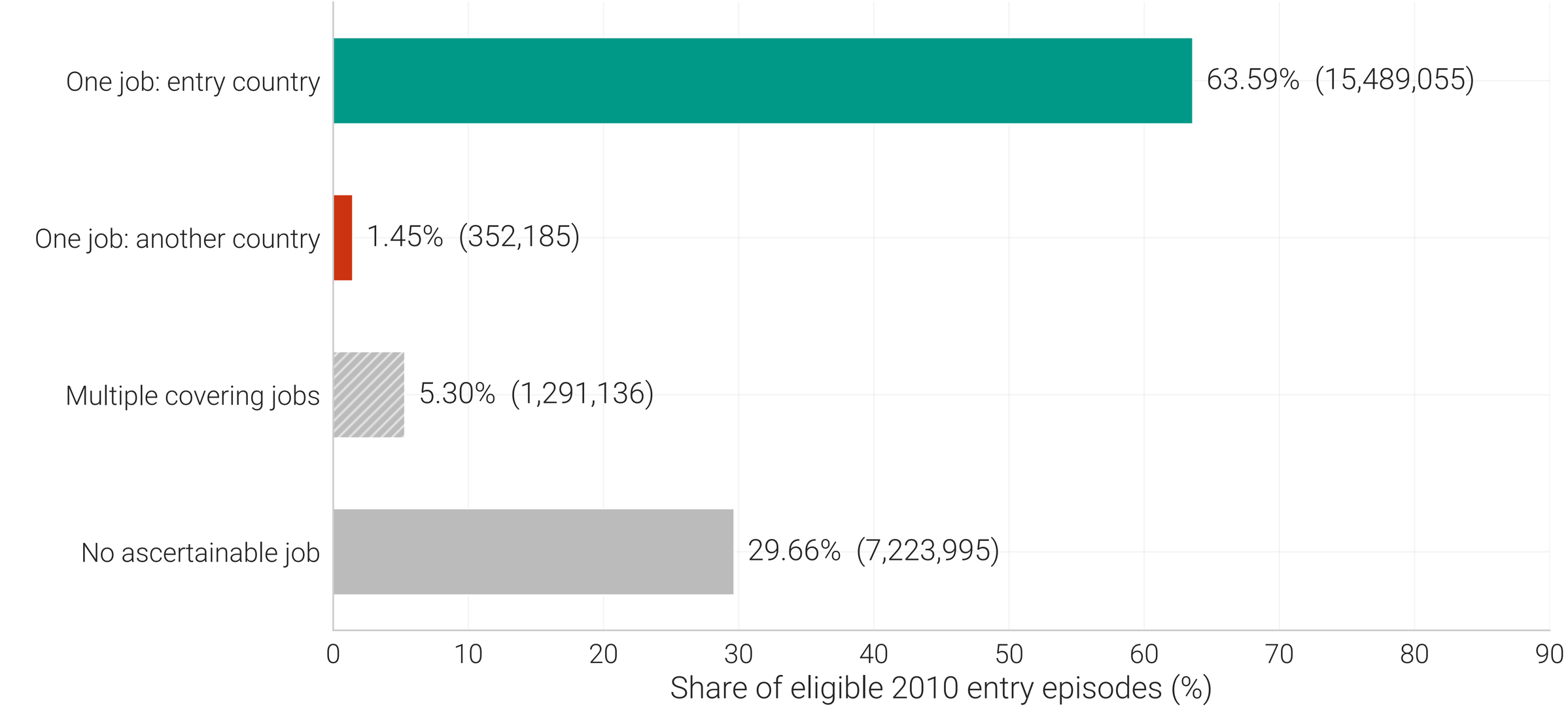
Measured positive adjacent-record gaps · not unemployment or legal-status gaps



▶ Return: histories

A fixed horizon requires a job covering the endpoint

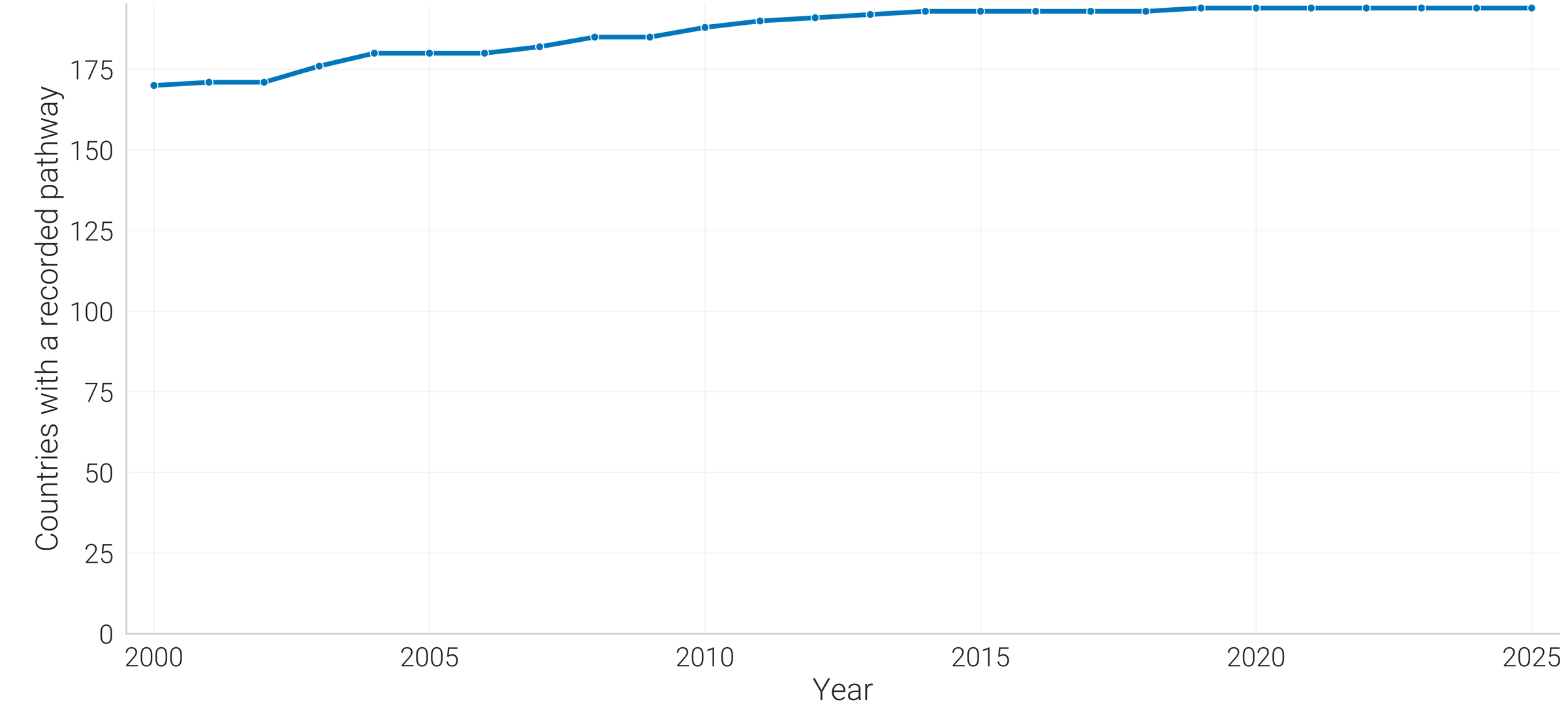
Corrected one-year endpoint comparison · unresolved follow-up is separate



▶ Return: histories

Historical representation broadens over time

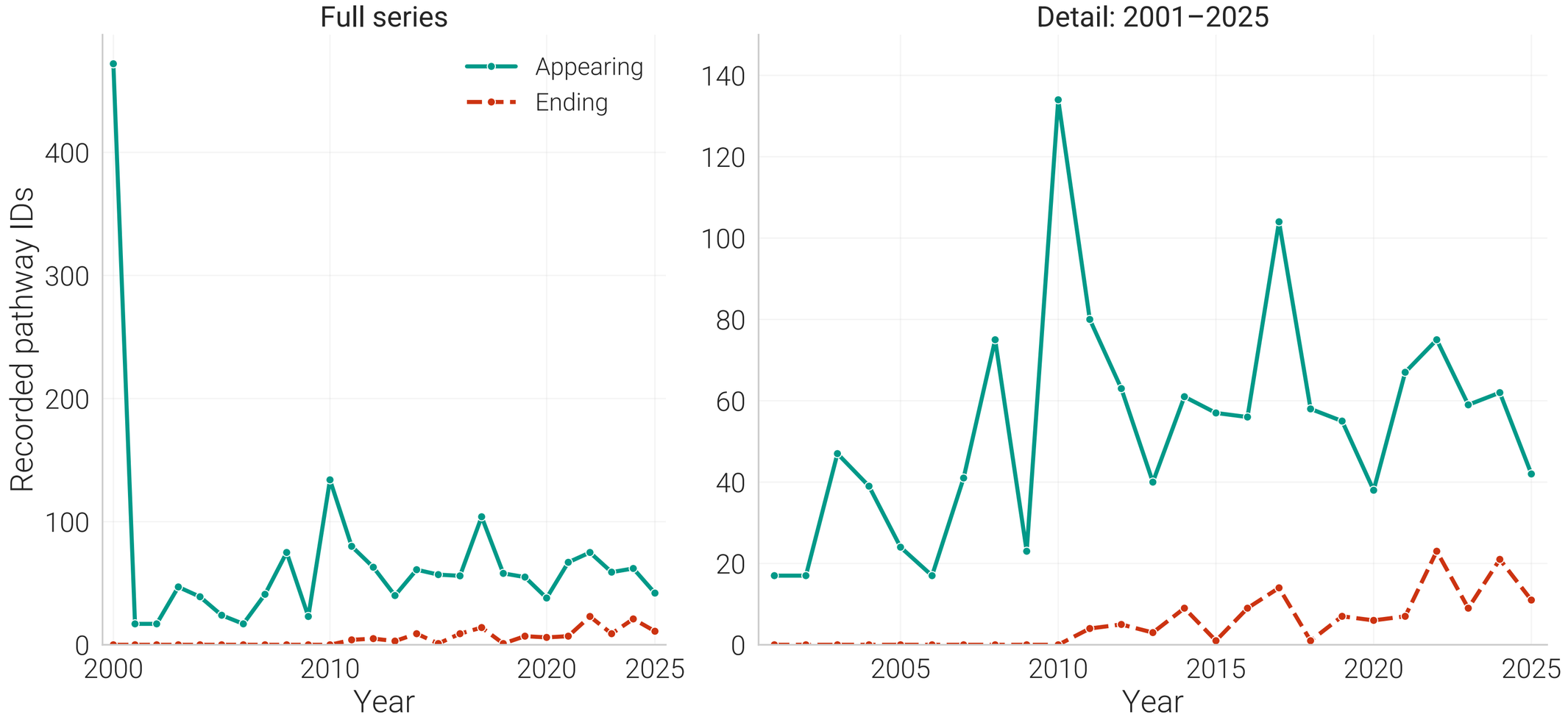
Represented destination labels · not necessarily new policy adopters



▶ Return: growth ▶ Return: bridge

Recorded appearances and endings are not legal births and deaths

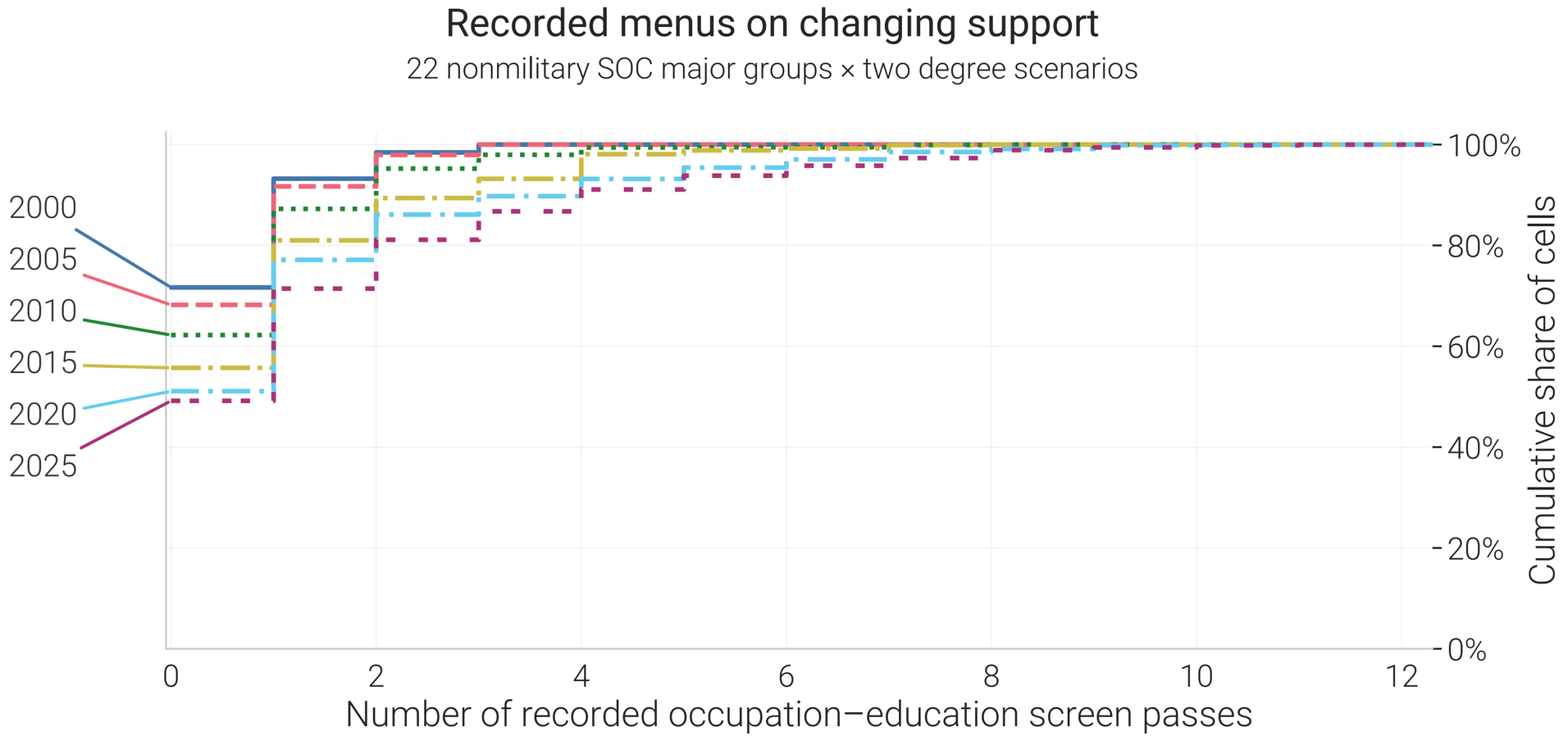
The 2000 boundary and unresolved status affect lifecycle interpretation



▶ Return: growth

Changing support gives a different pooled comparison

All represented destinations each year · 44 scenarios · six dates



Active destinations by year: 167/177/187/193/193/193; equal cell weights. Exact ECDF; no smoothing.

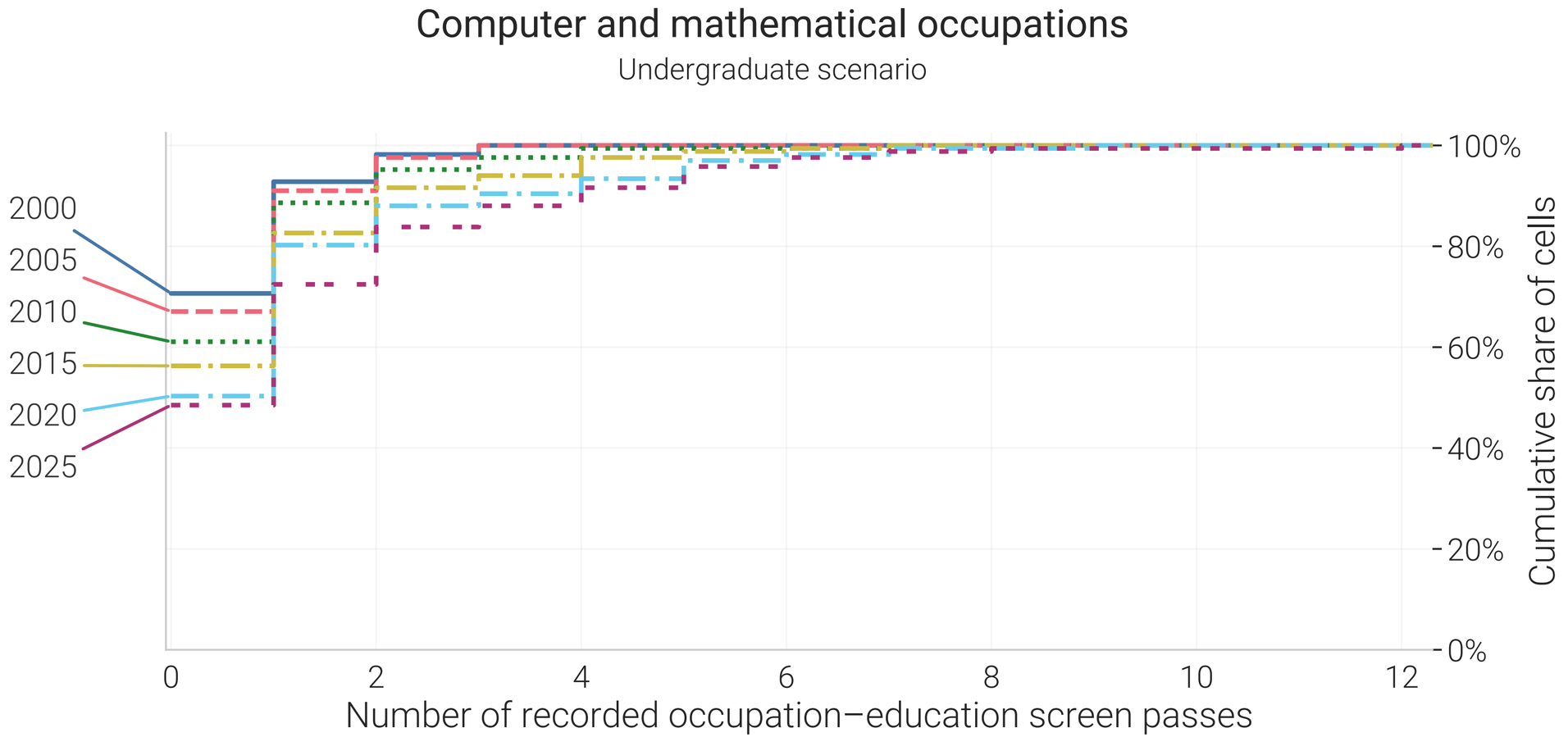
Zero coded passes are retained. Unknown screens are not zero legal access; nationality is not evaluated.

Retained hypothetical scenarios overlap; these are neither people nor complete legal choice sets.

▶ Return: fixed-support result

Computer/mathematical profile: fixed-support comparison

SOC 15 · undergraduate screen · six dates · no nationality or age screen



Fixed 167 destinations; equal cell weights. Exact ECDF; no smoothing.

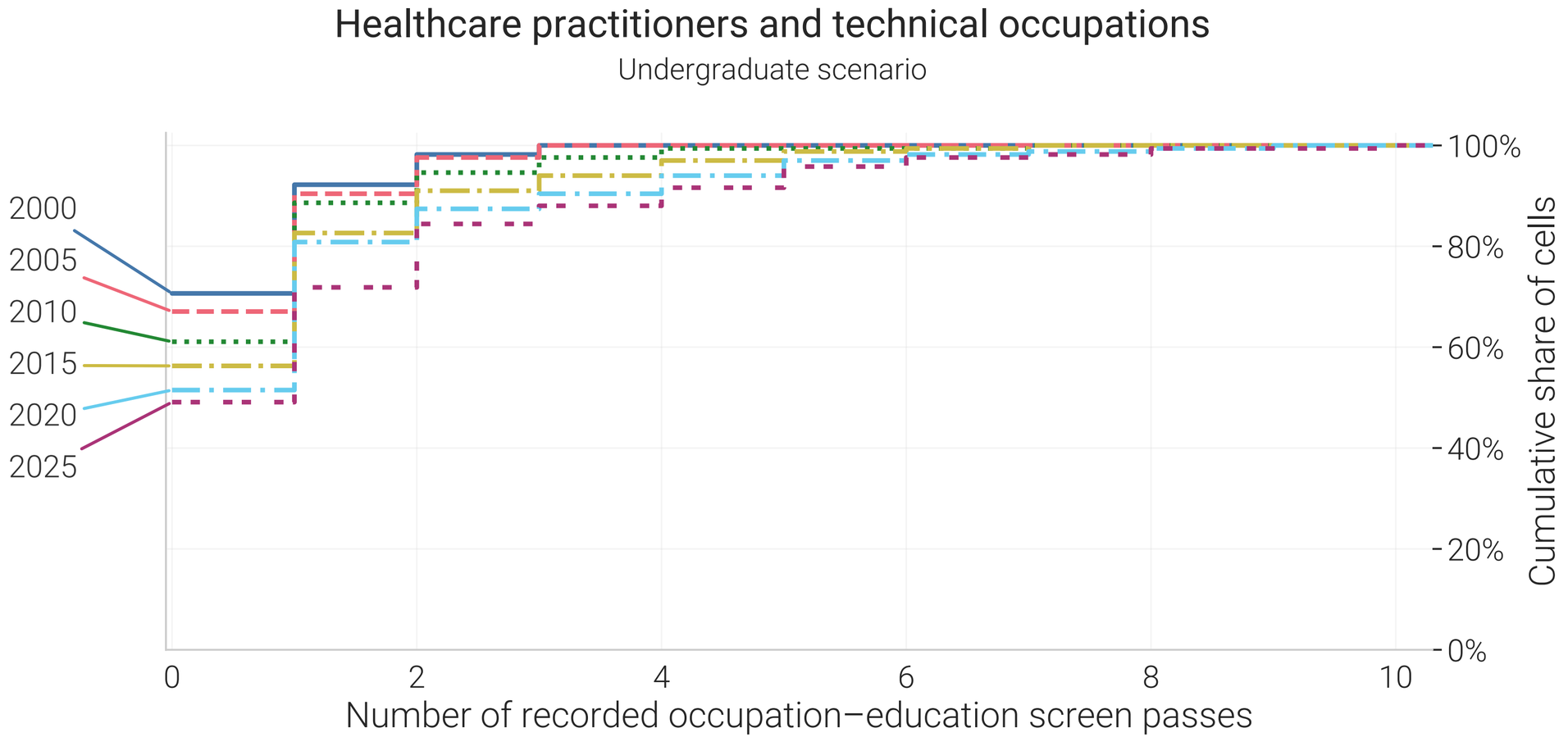
Zero coded passes are retained. Unknown screens are not zero legal access; nationality is not evaluated.

Retained hypothetical scenarios overlap; these are neither people nor complete legal choice sets.

▶ Return: pooled profiles

Healthcare-practitioner profile: fixed-support comparison

SOC 29 · undergraduate screen · six dates · no nationality or age screen



Fixed 167 destinations; equal cell weights. Exact ECDF; no smoothing.

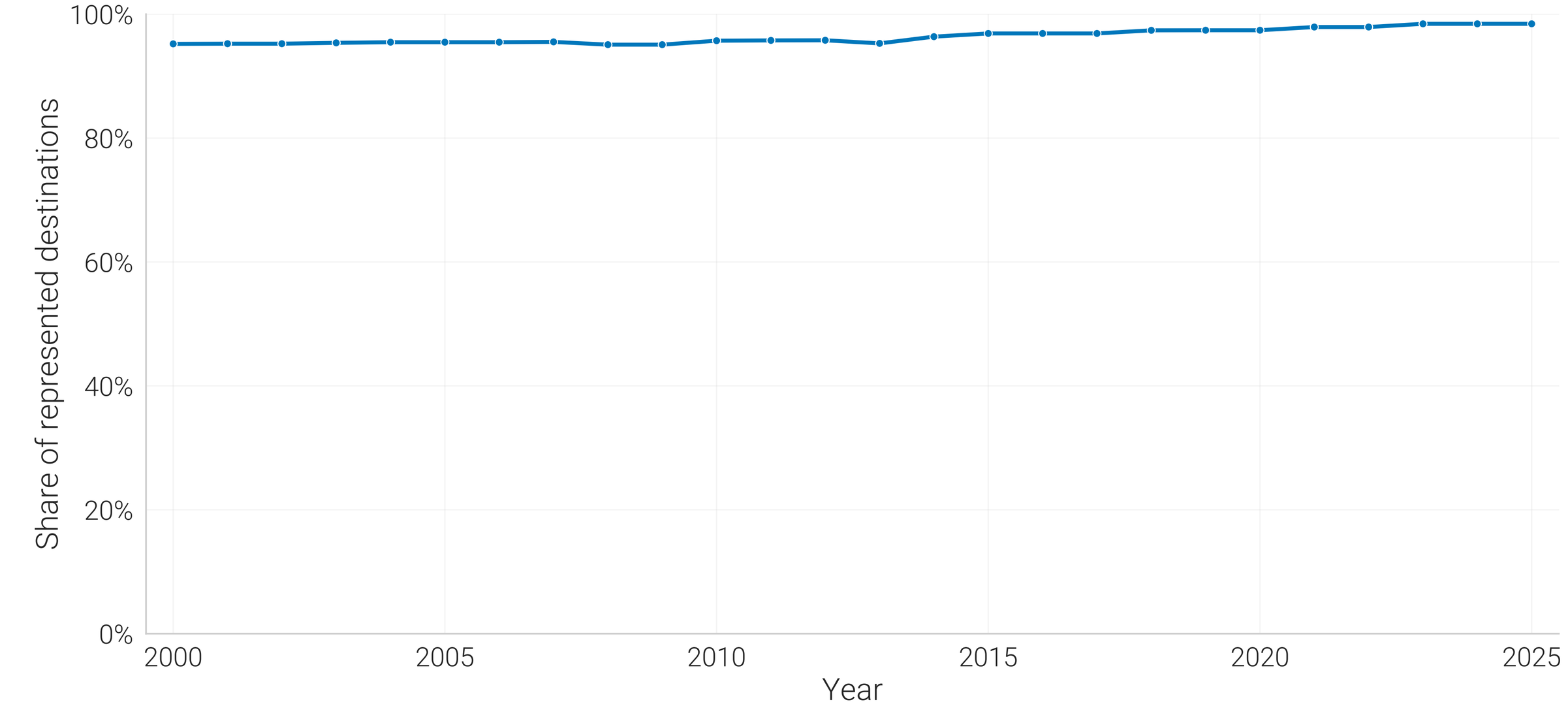
Zero coded passes are retained. Unknown screens are not zero legal access; nationality is not evaluated.

Retained hypothetical scenarios overlap; these are neither people nor complete legal choice sets.

▶ Return: pooled profiles

Unresolved rules prevent a complete legal-menu interpretation

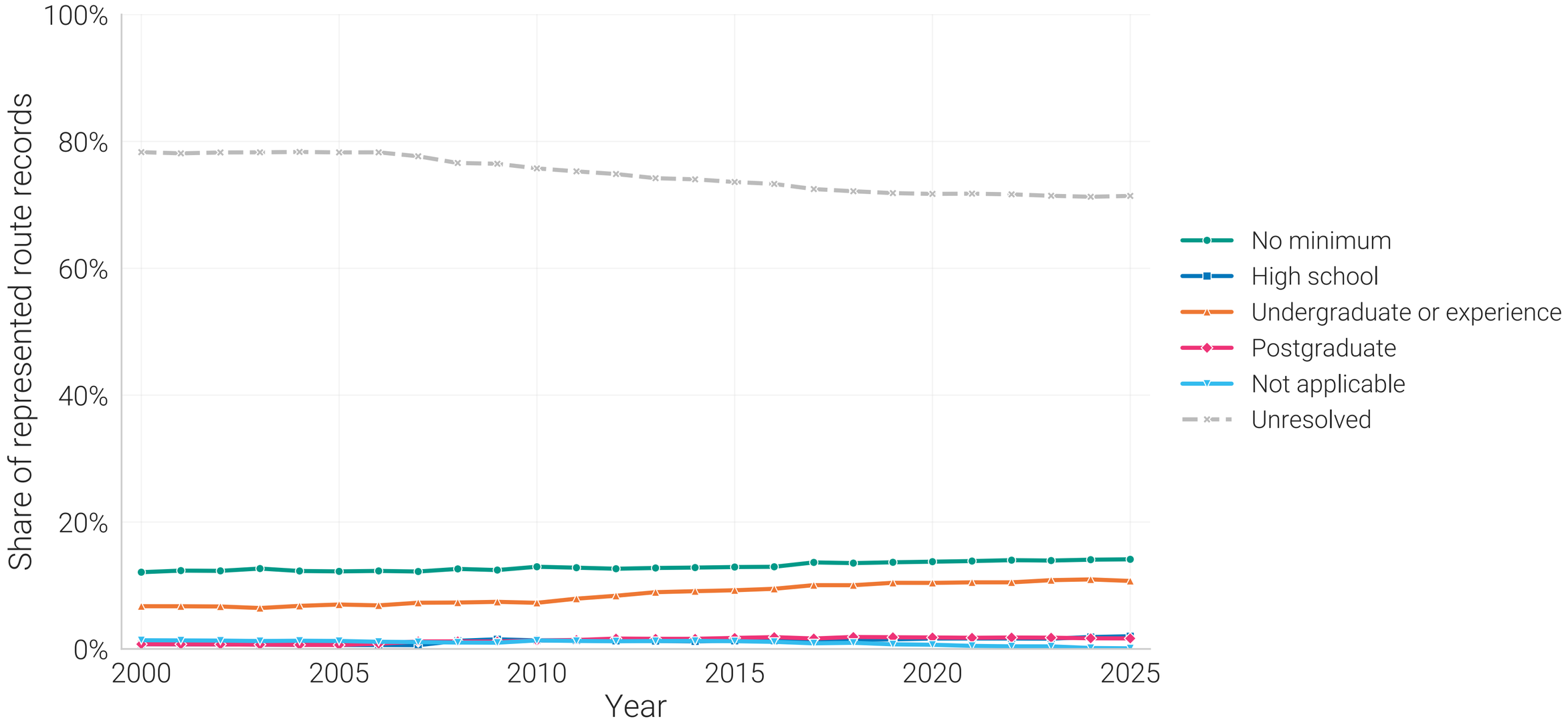
Recorded pass counts and unresolved evaluations are different objects



▶ Return: bridge ▶ Return: pooled profiles

Credential requirements are often unresolved

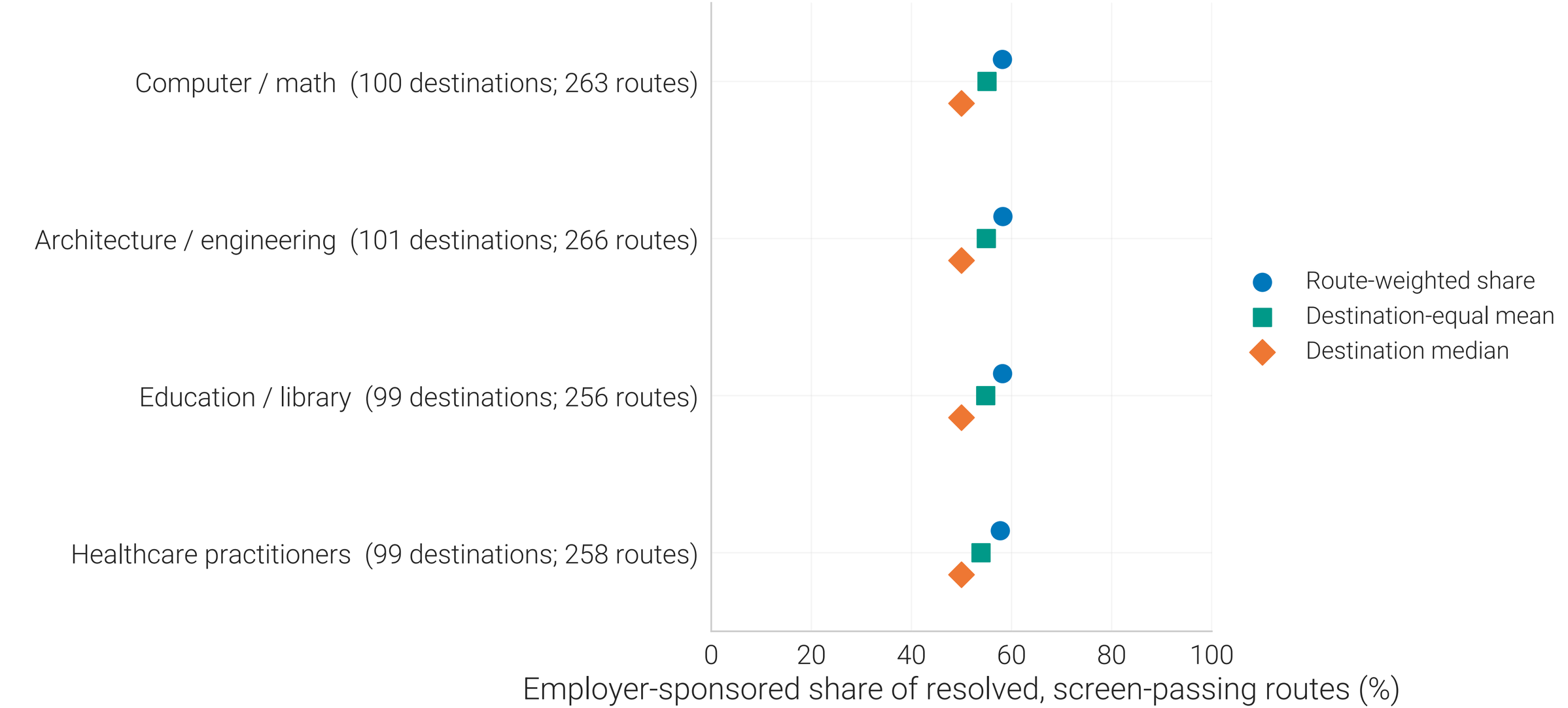
2000–2025 · represented route records · not an ordinal route-skill ranking



▶ Return: screening

Sponsorship shares depend on the weighting rule

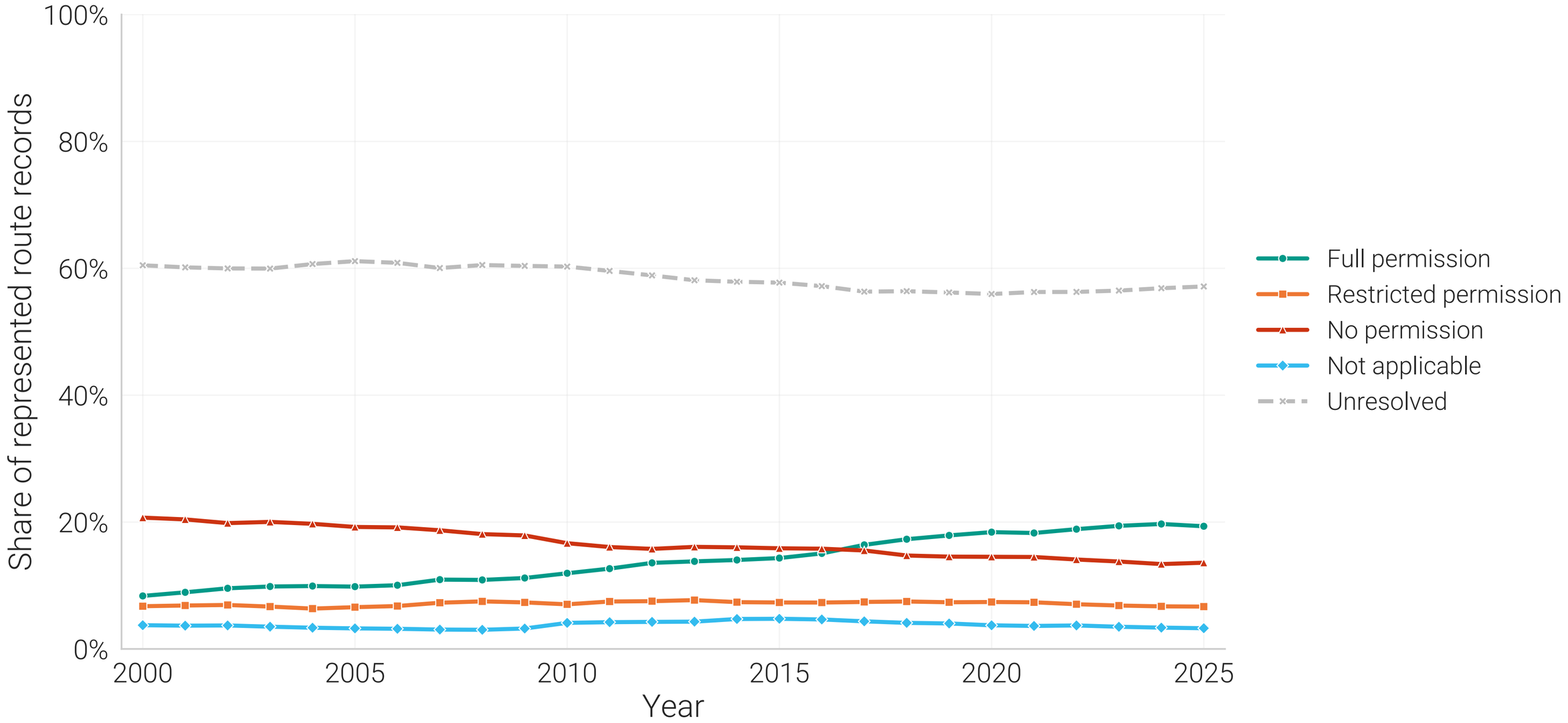
2024 · route-weighted, destination-equal and median comparisons



▶ Return: screening

Spouse work rights are one part of the route bundle

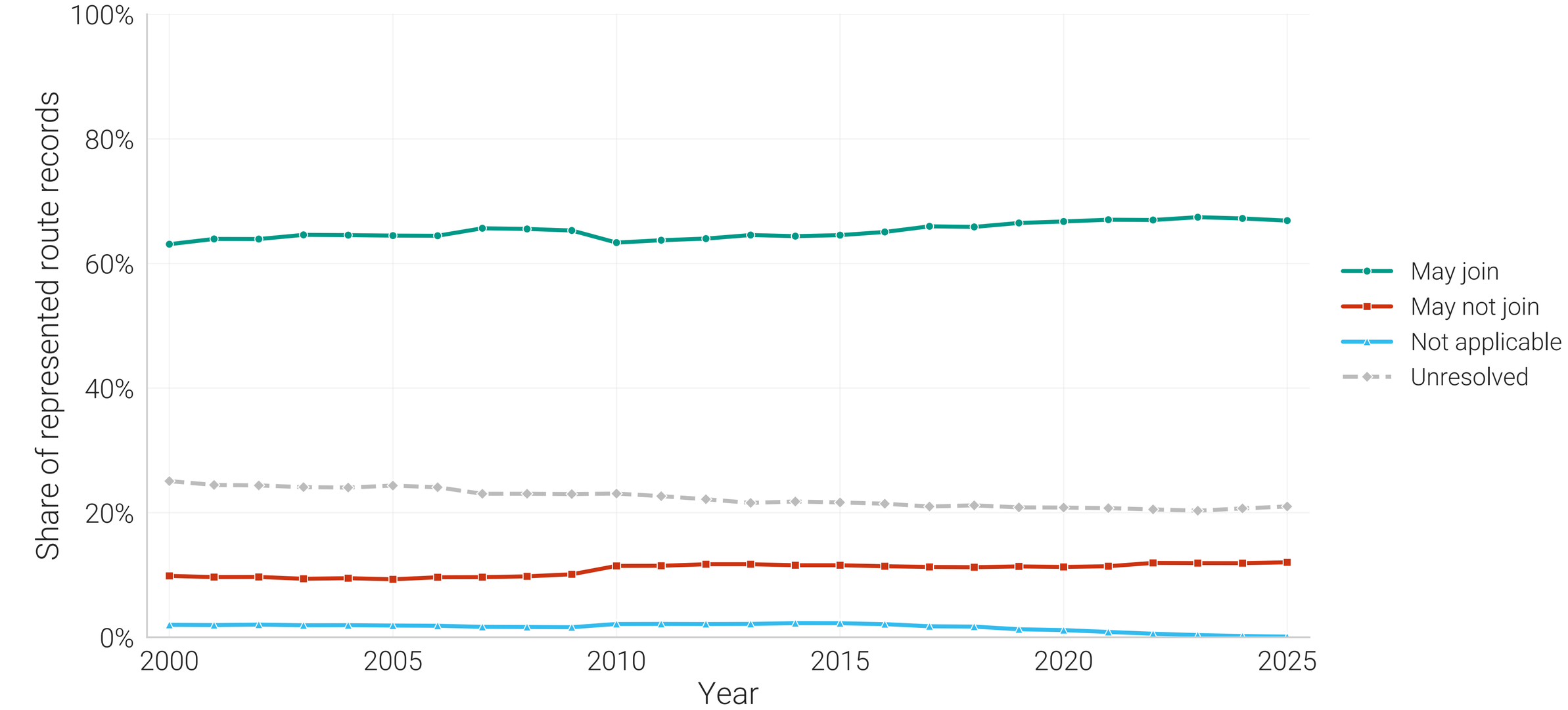
Represented route records · not an active-only or applicant-weighted denominator



▶ Dependent access ▶ Return: screening

Dependent access and dependent work rights are distinct

Represented route records · preserve unknown and inapplicable states

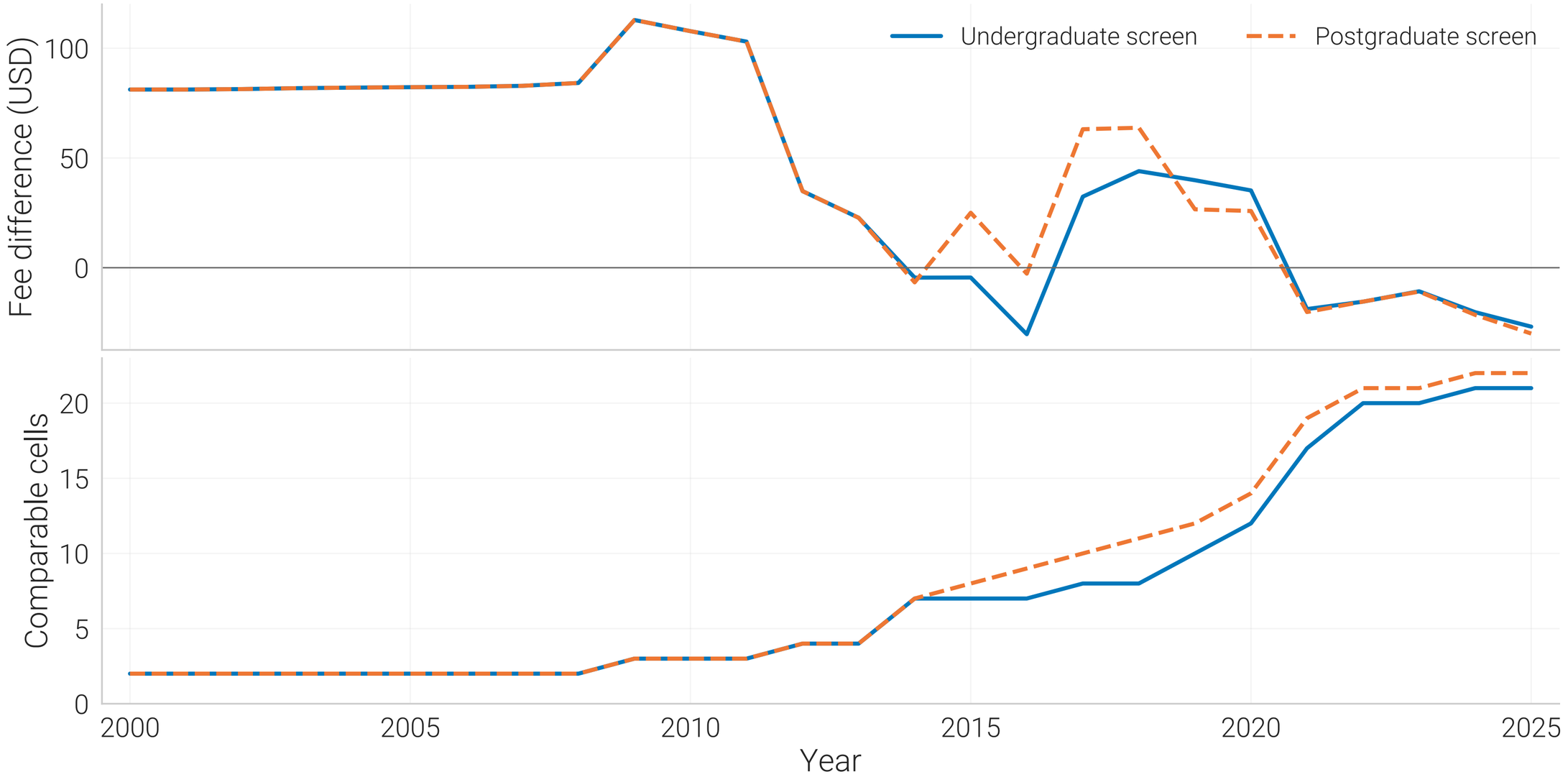


▶ Return: family-rights comparison

▶ Return: screening

Joint fee contrasts keep comparisons within a profile cell

SOC 15 · employer-sponsored minus explicit-none routes · comparable applicant fees



▶ Return: screening

Dominance needs complete, comparable bundles for one applicant

For declared dimensions, route A weakly dominates B only if A is weakly preferred on every dimension and strictly preferred on at least one.

- Fix citizenship, occupation, education, destination, year and application stage.
- Compare actual joint terms—not a synthetic combination of minima.
- Confirm the preference direction: a salary threshold is not a promised wage.
- Keep admission risk, sponsor availability, future rights and missing terms visible.

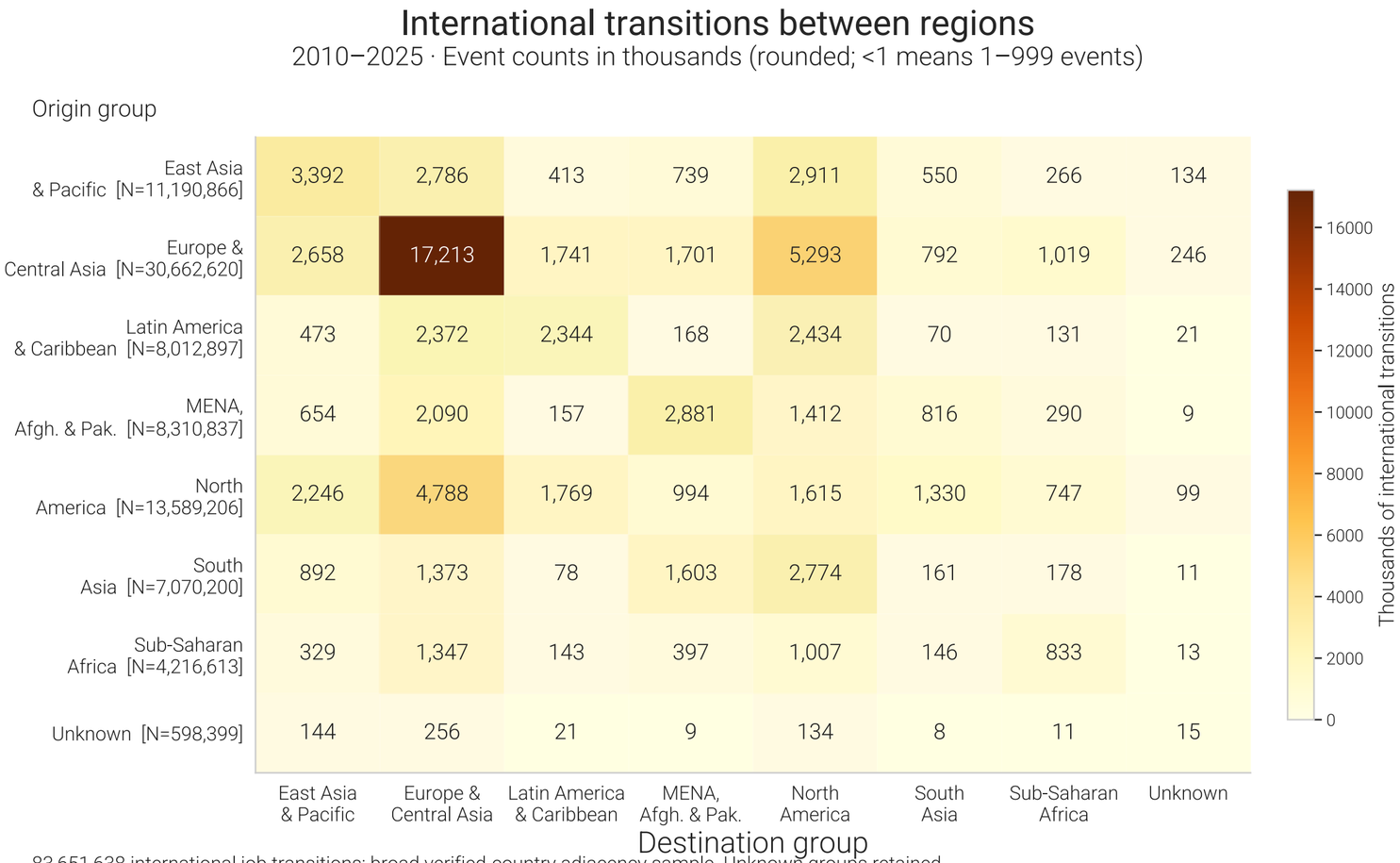
Report: dominated on observed dimensions; incomparable; or unassessable. Absence of observed dominance is not complete economic nondominance.

▶ Return: bundles

Appendix: group-to-group mobility

Region-to-region international transition volumes

2010–2025 · fixed World Bank 2026 regions · repeated international events



83,651,638 international job transitions; broad verified-country adjacency sample. Unknown groups retained.

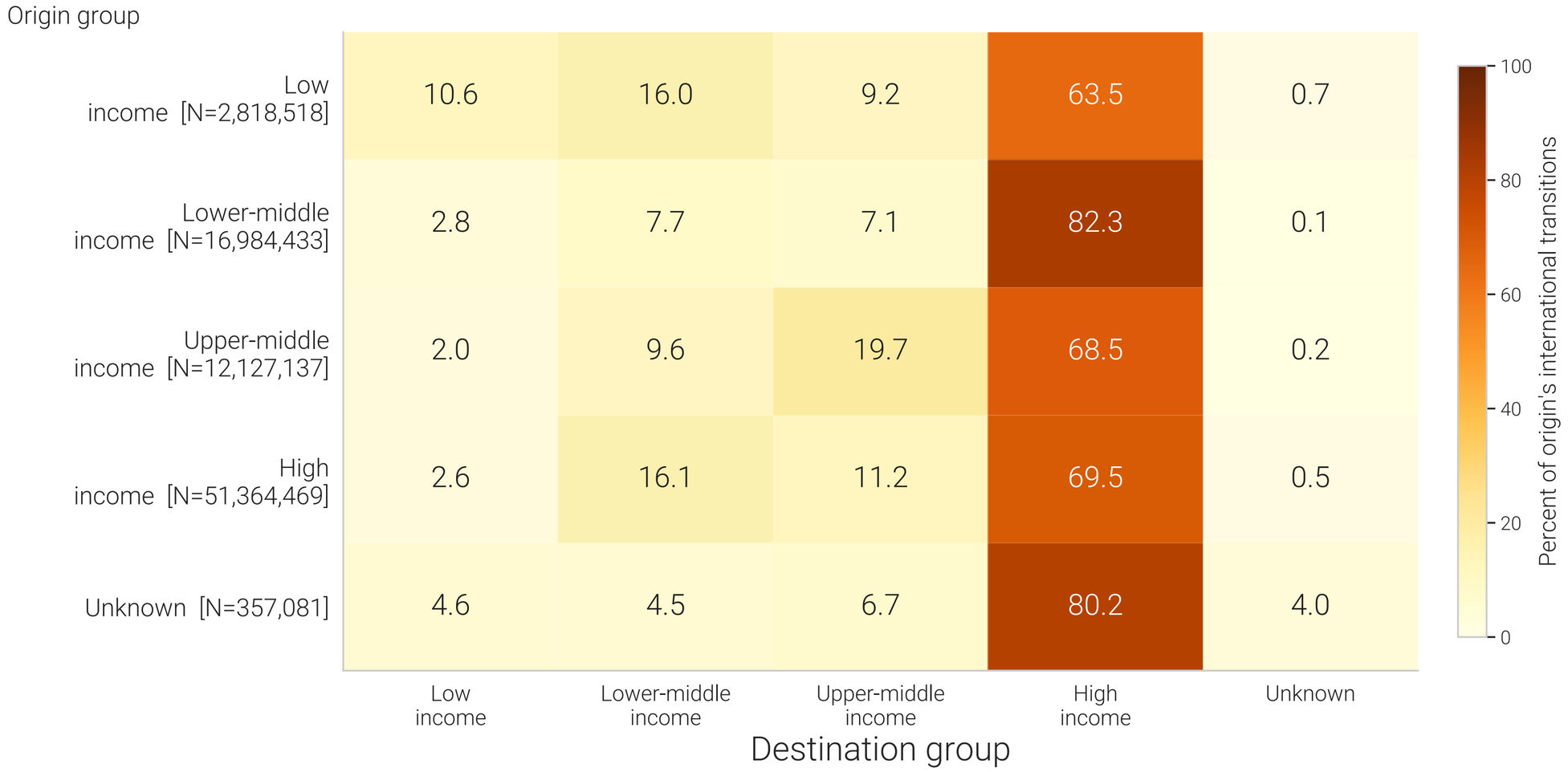
World Bank region metadata retrieved October 2026, held fixed across years; MENA includes Afghanistan and Pakistan.

Origin and destination refer to recorded jobs, not citizenship or verified residence. N is the origin row total.

▶ Return: origin-conditional shares

International transitions between income groups

2010–2025 · fixed World Bank FY2011 income groups · origin-conditional shares

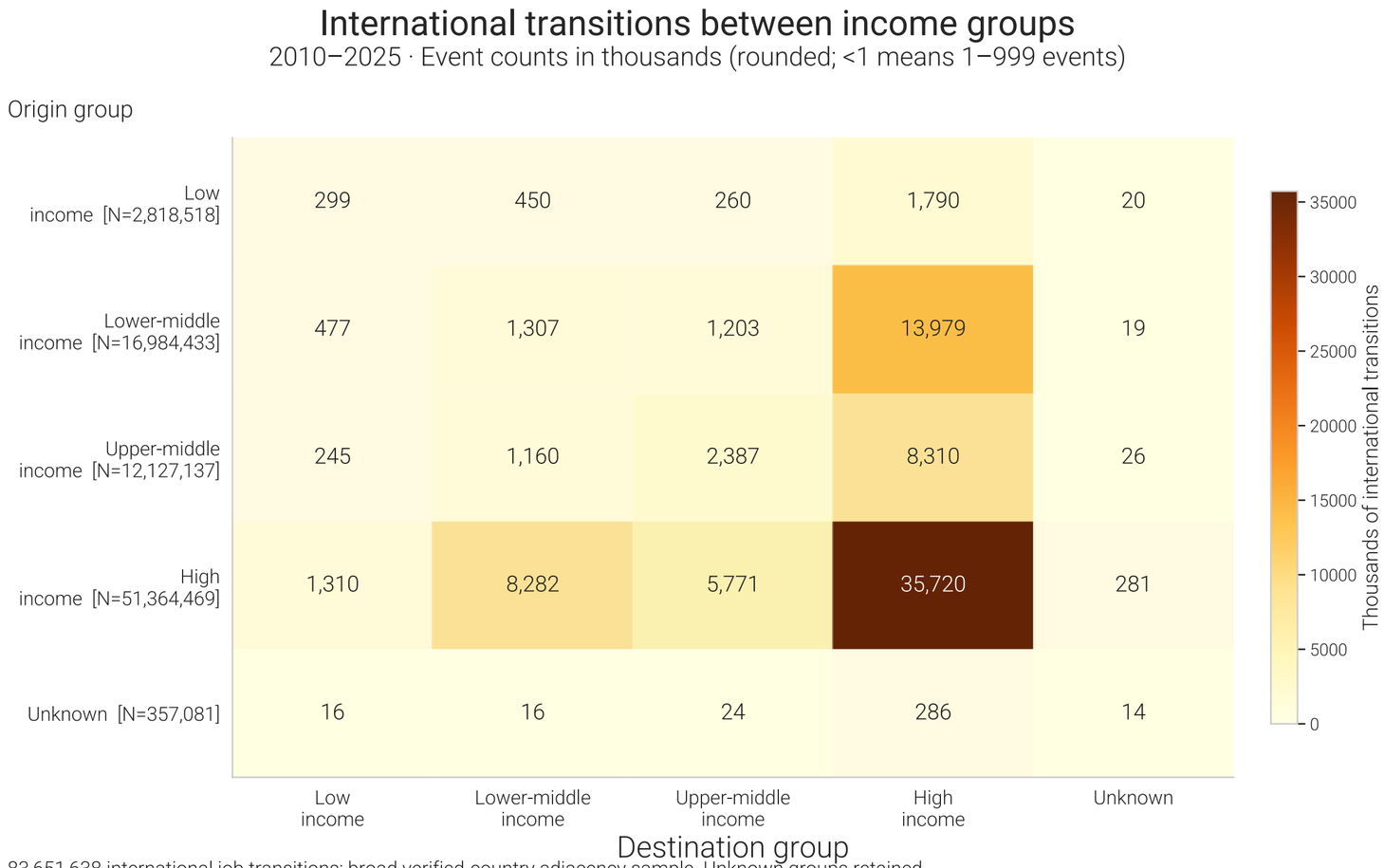


83.65m international job transitions · broad verified-country sample · unknown groups retained

▶ Counts ▶ Return: regional transitions

Income-group transition volumes reflect origin size too

2010–2025 · fixed World Bank FY2011 groups · country income, not personal wages



83,651,638 international job transitions; broad verified-country adjacency sample. Unknown groups retained.

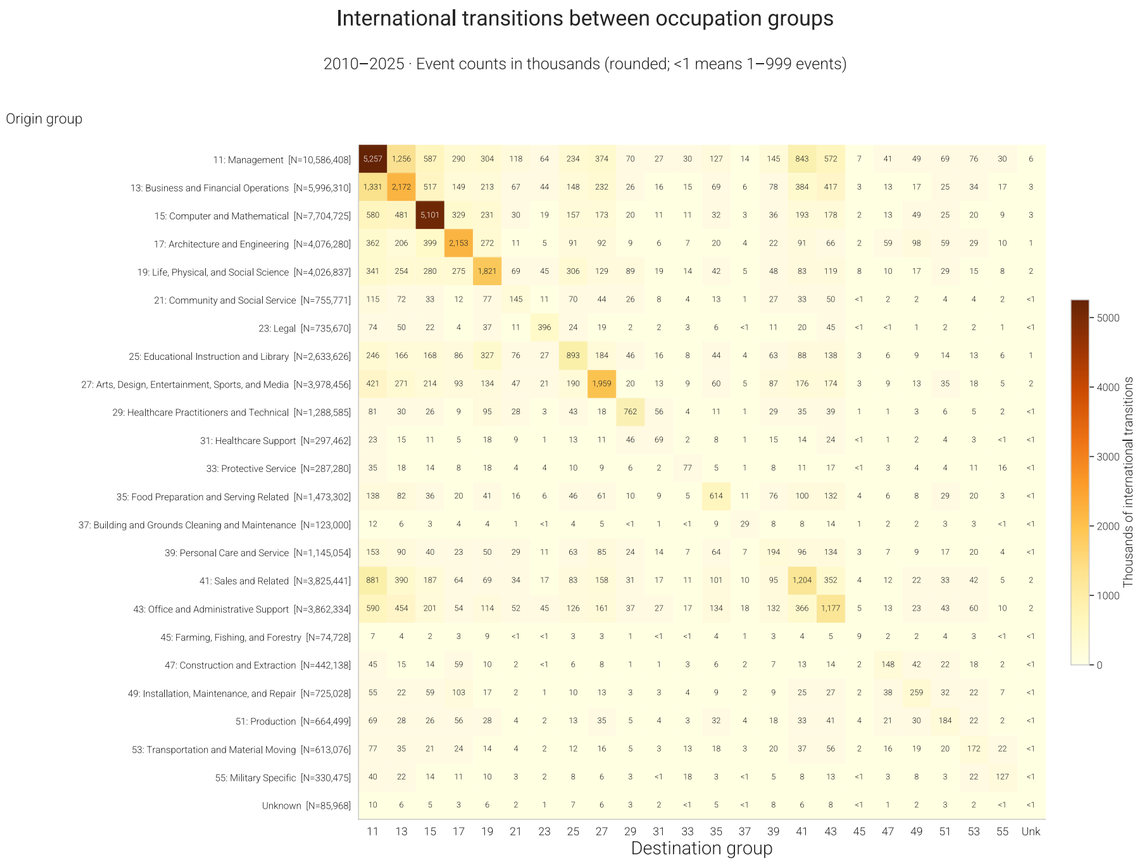
World Bank FY11 income classifications fixed across all years; income groups describe countries, not individual earnings.

Origin and destination refer to recorded jobs, not citizenship or verified residence. N is the origin row total.

▶ Income shares ▶ Return: regional transitions

Origin-to-destination occupation volumes

Recognised SOC major groups · international events · no ordinal skill interpretation



55,732,453 international job transitions; strict history sample (different from the geography sample). Unknown groups retained.

All 23 SOC major groups plus unknown; 113,932 events have an unknown endpoint occupation. This is not a skill ranking.

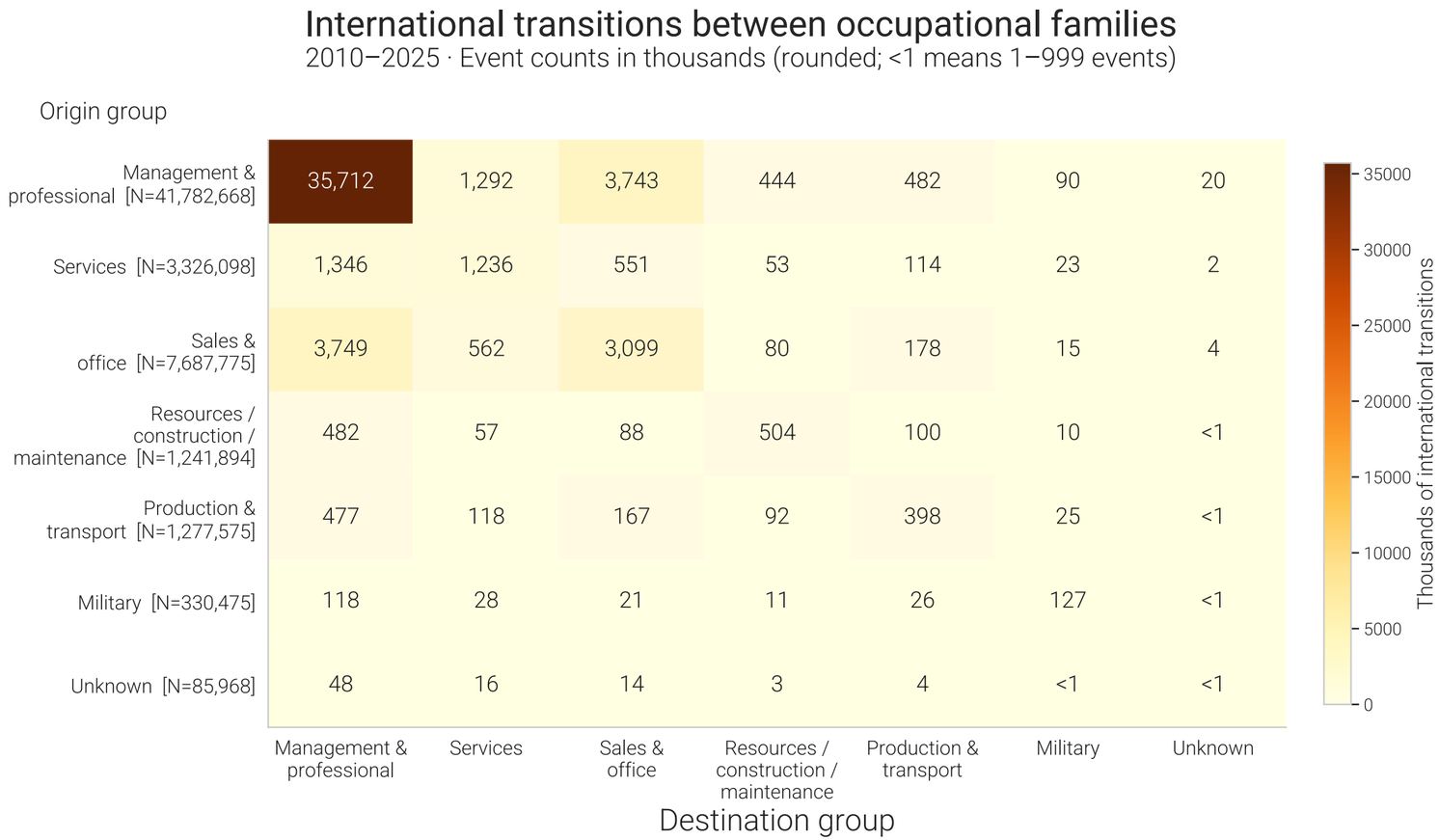
Origin and destination refer to recorded jobs, not citizenship or verified residence. N is the origin row total.

▶ Detailed occupation shares

▶ Return: occupational families

Broad occupational-family transition volumes

Six analytical families plus unknown · repeated international events



55,732,453 international job transitions; strict history sample; analytical families, not skill ranks. Unknown groups retained.

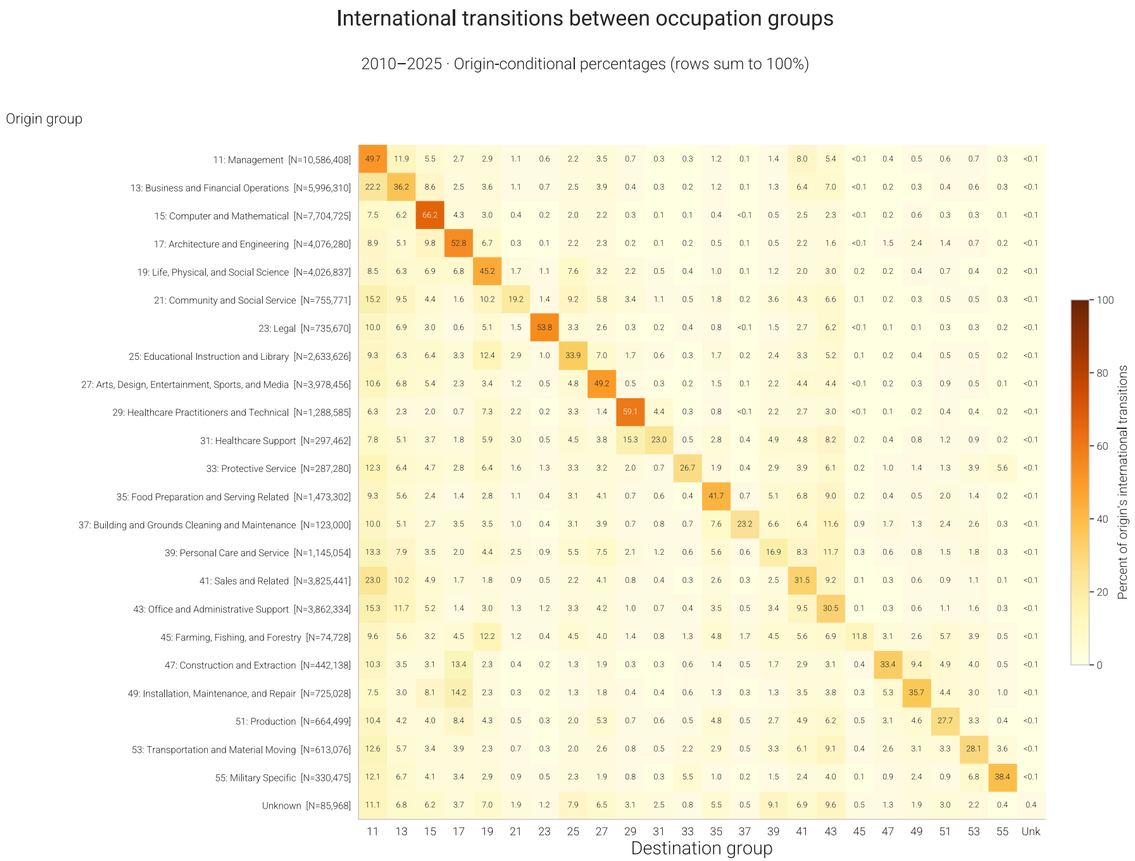
Families: 11–29; 31–39; 41/43; 45/47/49; 51/53; 55; unknown. Full SOC matrix retained separately.

Origin and destination refer to recorded jobs, not citizenship or verified residence. N is the origin row total.

▶ Return: family shares ▶ Detailed SOC counts

The full matrix retains every SOC major group

23 groups plus unknown · origin-conditional shares · detail for discussion and zoom



55,732,453 international job transitions; strict history sample (different from the geography sample). Unknown groups retained.

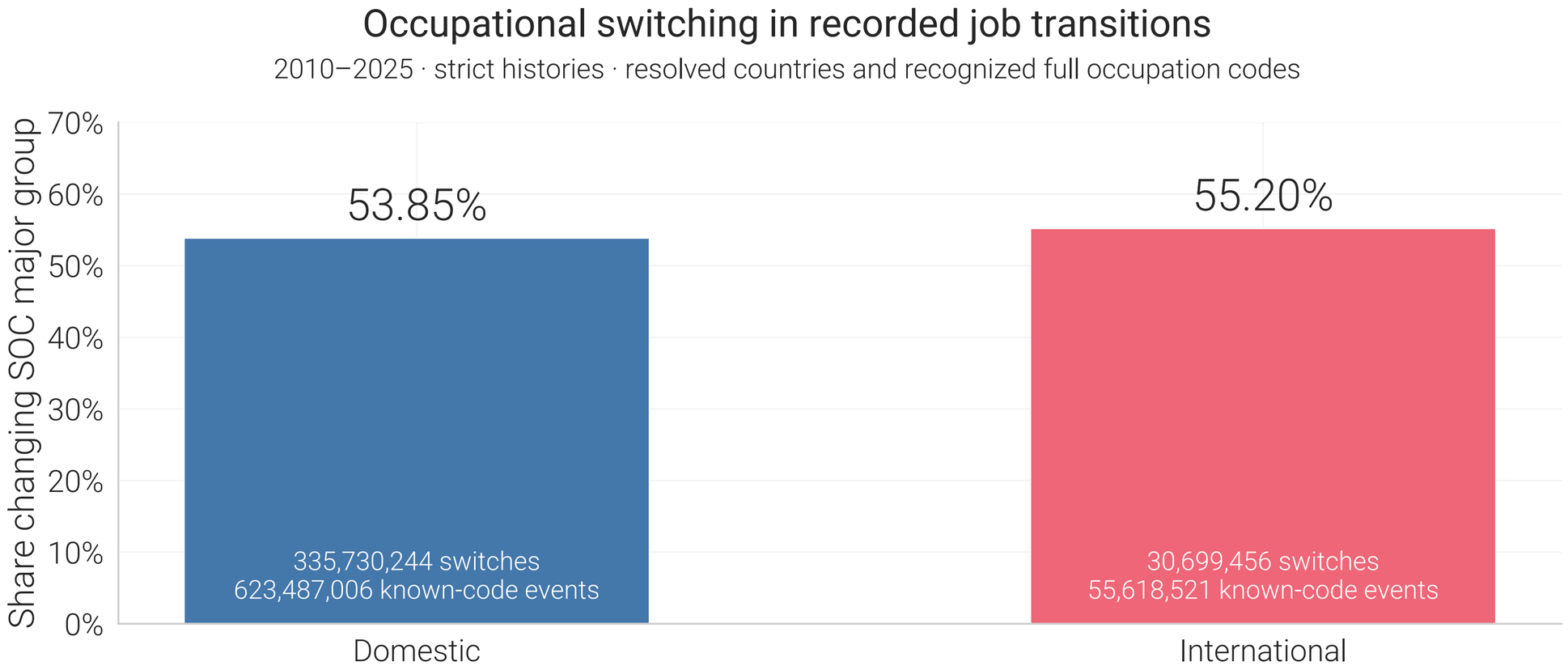
All 23 SOC major groups plus unknown; 113,932 events have an unknown endpoint occupation. This is not a skill ranking.

Origin and destination refer to recorded jobs, not citizenship or verified residence. N is the origin row total.

▶ Detailed counts ▶ Return: occupational families

Switching occurs in domestic and international job transitions

2010–2025 · recognised full occupation codes · event-weighted



Excluded unknown occupations: 1,850,579 domestic; 113,932 international. Country unresolved: 7,324.

Event-weighted, not person-weighted. SOC major-group changes do not establish skill upgrading or residence moves.

▶ Return: occupational transitions

Occupational groups are not a skill ladder

- SOC major groups describe kinds of work; their numeric codes are not ranks.
- The broad SOC-to-ISCO reference scenario permits upward, downward and equal changes for about **95%** of events.
- Routine/non-routine, cognitive/manual and preparation requirements are distinct concepts.
- A defensible skill-transition matrix needs a verified finer code/vintage and a declared crosswalk; it cannot be recovered from destination skill marginals.

The occupation matrix is descriptive evidence. A ranked skill matrix is not yet established.

▶ Return: occupational transitions ▶ Taxonomy

Appendix: estimation and literature

MPML normalises the supplied alternatives, not a missing population

$$\mu_{ra} = \exp(\alpha_r + \beta x_{ra} + z'_{ra}\gamma), \quad p_{ra} = \frac{\exp(\beta x_{ra} + z'_{ra}\gamma)}{\sum_{b \in A_r} \exp(\beta x_{rb} + z'_{rb}\gamma)}.$$

$$\ell(\beta, \gamma) = \sum_{r,a} Y_{ra} \log p_{ra}.$$

- Profiling the row effect gives the corresponding Poisson objective on **identical support and weights**.
- Current A_r contains observed positive, policy-matched international cells.
- Fitting supplied row totals does not recover unobserved stayers or unavailable destinations.

Fixed effects determine the monetary comparison

Specification	Conditional row	Additional fixed effects
L	Origin × year	Destination; occupation × education × year; dyad
D	Origin × year	Destination × year; occupation × education × year; dyad
T	Origin × occupation × education × year	Destination × year; dyad × occupation
K/E	Type-specific origins	Destination × occupation(× education) × year; dyad × occupation

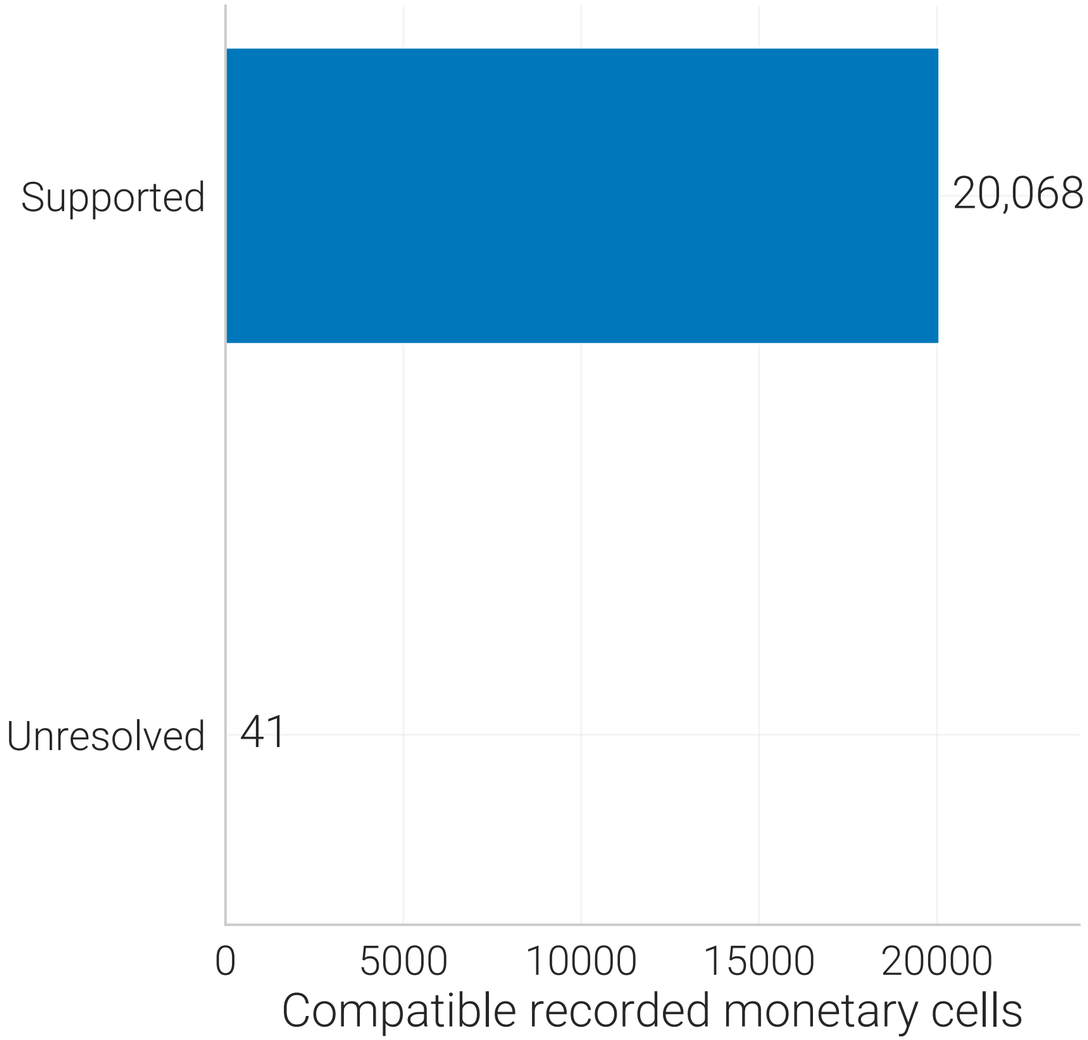
K/E absorb the monetary policy variation: **not identified**, not an estimated zero effect.

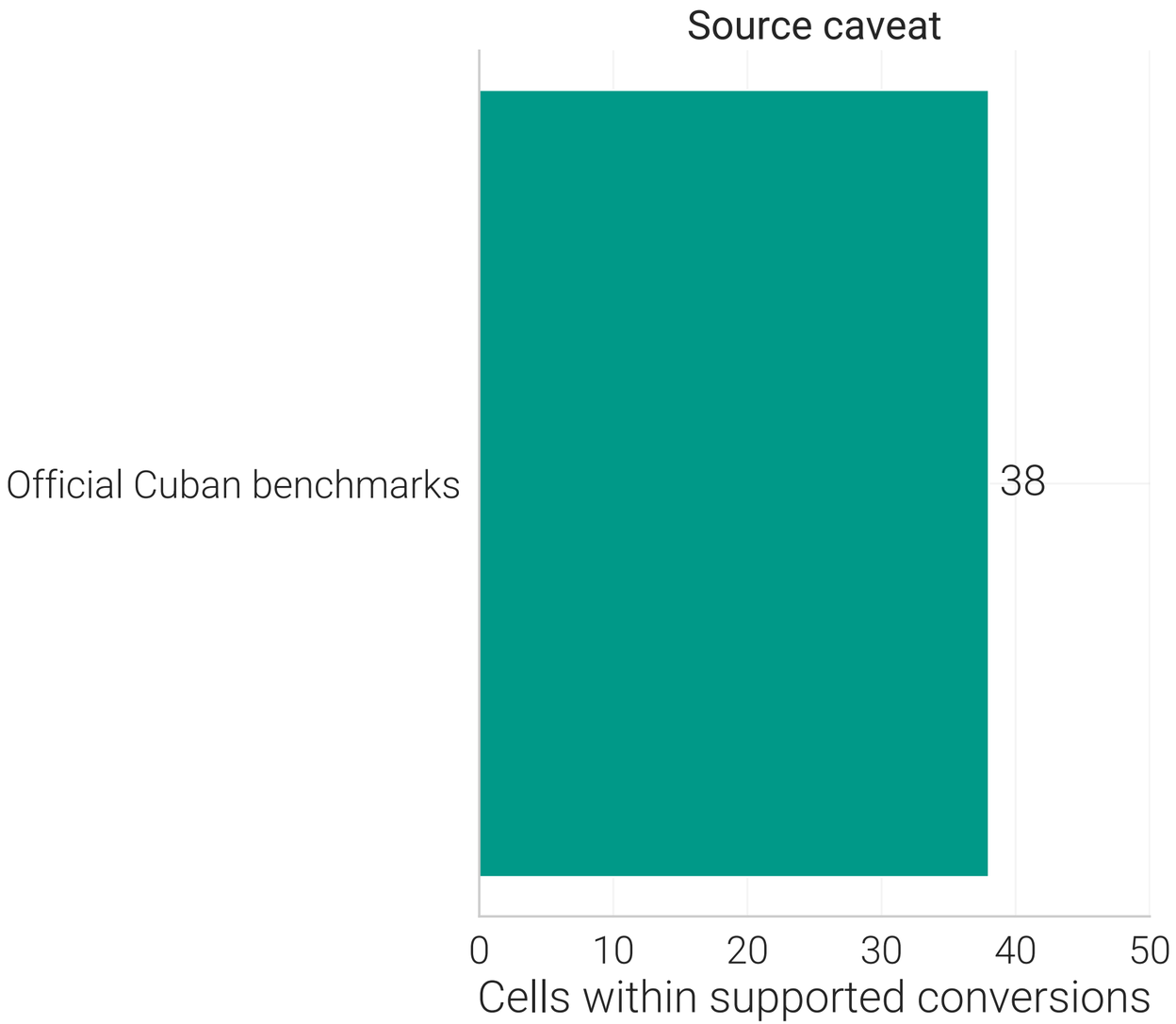
2010–2024; 12 origins, four SOC groups, two lifetime-education profiles. Fee: **121,258 cells / 7.52m events**; salary: **42,492 cells / 3.14m events**. Samples differ. No current standard errors.

▶ Return: worker responses ▶ Return: monetary preliminaries

Monetary comparisons need compatible amounts and units

FX support is not proof of fee definition, real value or incidence





▶ Return: monetary preliminaries

Rich histories do not by themselves separate preferences from access

(Berry and Haile 2024; Agarwal and Somaini 2026)

- Legal compatibility, consideration, sponsorship, admission and take-up are distinct stages.
- Observed characteristics and repeated choices add information; they do not eliminate market-level unobservables or supply excluded variation automatically.
- Applications, refusals, route-linked admissions and usable offers can help distinguish these stages.

Our task: define the choice occasion, population, support and policy variation before treating conditional allocation slopes as migrant elasticities.

▶ Return: contribution ▶ Return: worker responses

Fee competition identifies a net marginal object

$$G_d = f_d' n_d(f) + B_d(n_d) - C_d(n_d), \qquad c_d - b_d = f_d + (D_d^{\mathsf{T}})^{-1} n_d.$$

D_d is the own-route quantity-by-fee derivative matrix; b_d and c_d are marginal benefits and resource costs.

- Requires identified unilateral fee responses, matching fee/quantity units, interior optimality and no omitted binding constraint.
- Capacity adds a multiplier; application fees need application-stage revenue.
- Adding the same function to B_d and C_d leaves choices unchanged: unrestricted benefits and costs are **not separately identified**.

Trade-policy inversion provides a comparison, not a Visa identification theorem (Adão et al. 2025).

▶ Return: cost preliminaries ▶ Return: next steps

Welfare needs a population and an adjustment mechanism

- Migrants and families; origin residents; destination residents; employers; governments.
- Wages, amenities, fiscal effects and housing or other prices may adjust.
- Fees are transfers, waiting uses time, and restrictions can remove future options.
- Government recruitment gains do not establish gains for everyone affected.

Counterfactual scope: worker responses at fixed offers → market adjustment → government responses, adding assumptions at each step.

▶ Return: next steps

Policy measurement and worker allocation frame the contribution

- **Policy measurement:** levels, signed changes and instrument indicators have different units (Helbling et al. 2017; de Haas et al. 2018; Czaika and Parsons 2017).
- **Alternative destinations:** multilateral resistance and third-country diversion already have direct treatments (Bertoli and Fernández-Huertas Moraga 2013; Brinatti and Guo 2024).
- **Careers and equilibrium:** dynamic location choice; employer options and wage setting; connected domestic markets (Kennan and Walker 2011; Townsend and Allan 2025; Borusyak et al. 2026).
- **Identification and policy:** constrained demand and revealed policy objectives require setting-specific restrictions (Berry and Haile 2024; Agarwal and Somaini 2026; Adão et al. 2025).

Proposed contribution: connect dated legal bundles and observed histories to cross-country allocation and, with further identification, competing government offers.

▶ Return: contribution ▶ References

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▶ Return: contribution ▶ Return: conclusion